

OPAx355 200-MHz CMOS Operational Amplifiers With Shutdown

1 Features

- Unity-Gain Bandwidth: 450 MHz
- Wide Bandwidth: 200 MHz GBW
- Low Noise: 5.8 nV/√Hz
- Excellent Video Performance
 - Differential Gain: 0.02%
 - Differential Phase: 0.05°
 - 0.1-dB Gain Flatness: 75 MHz
- Input Range Includes Ground
- Rail-to-Rail Output (within 100 mV)
- Low Input Bias Current: 3 pA
- Low Shutdown Current: 3.4 μA
- Enable and Disable Time: 100 ns and 30 ns
- Thermal Shutdown
- Single-Supply Operating Range: 2.5 V to 5.5 V
- MicroSIZE Packages

2 Applications

- Video Processing
- Ultrasound
- Optical Networking, Tunable Lasers
- **Photodiode Transimpedance Amplifiers**
- Active Filters
- High-Speed Integrators
- Analog-to-Digital Converter (ADC) Input Buffers
- Digital-to-Analog Converter (DAC) Output Amplifiers
- Barcode Scanners
- Communications

3 Description

The OPA355 series of high-speed, voltage-feedback CMOS operational amplifiers are designed for video and other applications requiring wide bandwidth. The OPA355 series is unity-gain stable and can drive large output currents. In addition, the OPAx355 series has a digital shutdown (enable) function. This feature provides power saving during idle periods and places the output in a high-impedance state to support output multiplexing. The differential gain is 0.02% and the differential phase is 0.05°. The quiescent current is 8.3 mA per channel.

The OPAx355 series is optimized for operation on single supply or dual supplies as low as 2.5 V (± 1.25 V) and up to 5.5 V (± 2.75 V). The common-mode input range for the OPAx355 series extends 100 mV below ground and up to 1.5 V from V+. The output swing is within 100 mV of the rails, supporting wide dynamic range.

The OPAx355 series is available in single (SOT-23-6 and SO-8), dual (VSSOP-10), and triple (TSSOP-14 and SO-14) versions. Multichannel versions feature completely independent circuitry for lowest crosstalk and freedom from interaction. All packages are specified from –40°C to +125°C.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
OPA355	SOIC (8)	4.90 mm × 3.91 mm
	SOT-23 (6)	2.90 mm × 1.60 mm
OPA2355	VSSOP (10)	3.00 mm × 3.00 mm
OPA3355	SOIC (14)	8.65 mm × 3.91 mm
	TSSOP (14)	5.00 mm × 4.40 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic

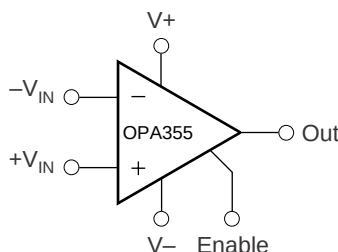


Table of Contents

1 Features	1	8.3 Feature Description	16
2 Applications	1	8.4 Device Functional Modes	17
3 Description	1	9 Application and Implementation	18
4 Revision History	2	9.1 Application Information	18
5 Device Comparison Table	3	9.2 Typical Applications	18
6 Pin Configuration and Functions	4	9.3 Video	22
7 Specifications	7	9.4 Wideband Video Multiplexing	23
7.1 Absolute Maximum Ratings	7	10 Power Supply Recommendations	25
7.2 ESD Ratings	7	11 Layout	25
7.3 Recommended Operating Conditions	7	11.1 Layout Guidelines	25
7.4 Thermal Information: OPA355	8	11.2 Layout Example	25
7.5 Thermal Information: OPA2355	8	12 Device and Documentation Support	26
7.6 Thermal Information: OPA3355	8	12.1 Related Links	26
7.7 Electrical Characteristics: $V_S = 2.7\text{ V to }5.5\text{ V}$ (Single-Supply)	9	12.2 Receiving Notification of Documentation Updates	26
7.8 Typical Characteristics	11	12.3 Community Resources	26
8 Detailed Description	16	12.4 Trademarks	26
8.1 Overview	16	12.5 Electrostatic Discharge Caution	26
8.2 Functional Block Diagram	16	12.6 Glossary	26
		13 Mechanical, Packaging, and Orderable Information	27

4 Revision History

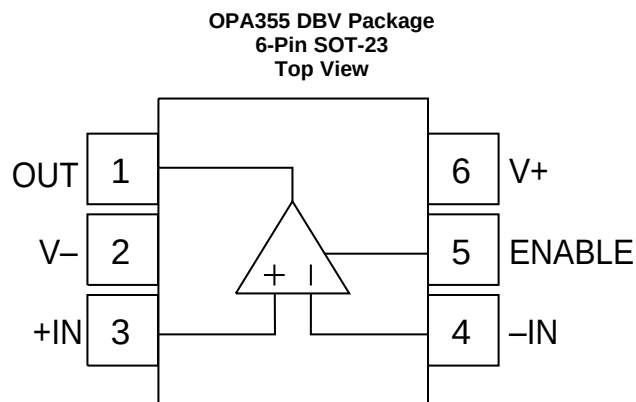
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision D (January 2004) to Revision E	Page
• Updated data sheet to latest TIS documentation and translation standards	1
• Changed pin type typo from MSOP to VSSOP in <i>Description</i> section	1
• Added <i>Device Information</i> table	1
• Deleted the <i>Absolute Maximum Ratings</i> table note: Input terminals are diode-clamped to the power-supply rails. Input signals that can swing more than 0.5 V beyond the supply rails should be current limited to 10 mA or less.	7
• Added <i>ESD Ratings</i> table	7
• Added <i>Recommended Operating Conditions</i> table	7
• Added <i>Thermal Information</i> tables	8
• Changed pin type typo from MSOP to VSSOP in <i>Electrical Characteristics</i> section	10
• Deleted the test conditions statement from <i>Typical Characteristics</i> graphs and moved the conditions to tablenotes below the graphs	11
• Added <i>Detailed Description</i> section	16
• Added <i>Functional Block Diagram</i> graphic	16
• Deleted <i>Input and ESD Protection</i> section	16
• Added <i>Application and Implementation</i> section	18
• Deleted <i>Internal ESD Protection</i> application	18
• Added <i>Power Supply Recommendations</i> section	25
• Added <i>Layout Guidelines</i> section	25

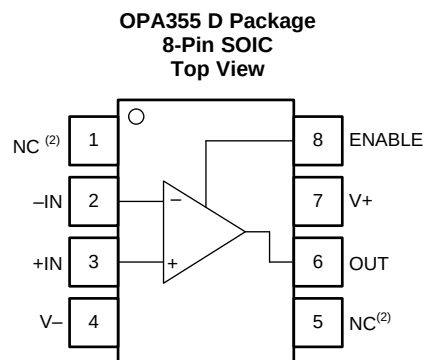
5 Device Comparison Table

OPAx355 RELATED PRODUCTS	FEATURES
OPA356	200-MHz, Rail-to-Rail Output, CMOS, No Shutdown
OPAx350	38-MHz, Rail-to-Rail Input and Output, CMOS
OPAx631	75-MHz, Rail-to-Rail Output
OPAx634	150-MHz, Rail-to-Rail Output
THS412x	Differential Input and Output, 3.3-V Supply

6 Pin Configuration and Functions



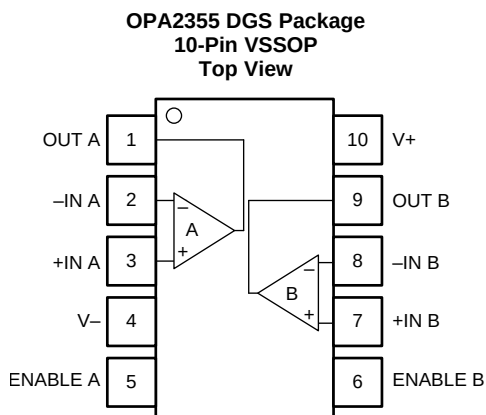
- (1) Pin 1 of the SOT-23-6 is determined by orienting the package marking as indicated in the diagram.



- (1) Pin 1 of the SOT-23-6 is determined by orienting the package marking as indicated in the diagram.
- (2) NC - no internal connection

Pin Functions: OPA355

NAME	PIN		I/O	DESCRIPTION
	SOT-23	SOIC		
ENABLE	5	8	—	Amplifier power down. Low = disabled, high = normal operation (pin must be driven)
IN+	3	3	I	Noninverting input pin
IN–	4	2	I	Inverting input pin
NC	—	1,5	—	Do not connect.
OUT	1	6	O	Output pin
V+	6	7	—	Positive power supply
V–	2	4	—	Negative power supply

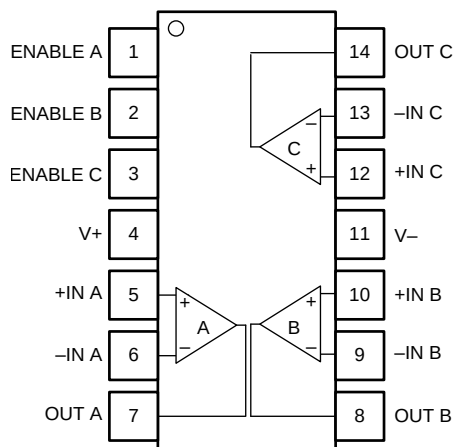


Pin Functions: OPA2355

PIN		I/O	DESCRIPTION
NAME	NO.		
ENABLE A	5	—	Amplifier power down, channel A. Low = disabled, high = normal operation (pin must be driven)
ENABLE B	6	—	Amplifier power down, channel B. Low = disabled, high = normal operation (pin must be driven)
+IN A	3	I	Noninverting input pin, channel A
+IN B	7	I	Noninverting input pin, channel B
–IN A	2	I	Inverting input pin, channel A
–IN B	8	I	Inverting input pin, channel B
OUT A	1	O	Output pin, channel A
OUT B	9	O	Output pin, channel B
V+	10	—	Positive power supply
V–	4	—	Negative power supply

OPA355, OPA2355, OPA3355

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**OPA3355 D and PW Packages
14-Pin SOIC, TSSOP
Top View**

Pin Functions: OPA3355

PIN		I/O	DESCRIPTION
NAME	NO.		
ENABLE A	1	—	Amplifier power down, channel A. Low = disabled, high = normal operation (pin must be driven)
ENABLE B	2	—	Amplifier power down, channel B. Low = disabled, high = normal operation (pin must be driven)
ENABLE C	3	—	Amplifier power down, channel C. Low = disabled, high = normal operation (pin must be driven)
+IN A	5	I	Noninverting input pin, channel A
+IN B	10	I	Noninverting input pin, channel B
+IN C	12	I	Noninverting input pin, channel C
–IN A	6	I	Inverting input pin, channel A
–IN B	9	I	Inverting input pin, channel B
–IN C	13	I	Inverting input pin, channel C
OUT A	7	O	Output, channel A
OUT B	8	O	Output channel B
OUT C	14	O	Output, channel C
V+	4	—	Positive power supply
V–	11	—	Negative power supply

7 Specifications

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Supply voltage	V+ to V–		7.5	V
Signal input terminals	Voltage	(V–) – 0.5	(V+) + 0.5	V
	Current		10	mA
Output short circuit ⁽²⁾		Continuous		
Operating temperature		–55	150	°C
Junction temperature			160	°C
Lead temperature (soldering, 10 seconds)			300	°C
Storage temperature range, T _{stg}		–65	150	°C

- (1) Stresses above *Absolute Maximum Ratings* may cause permanent damage. Exposure to absolute maximum conditions for extended periods may degrade device reliability. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those specified is not implied.

- (2) Short-circuit to ground, one amplifier per package.

7.2 ESD Ratings

		VALUE	UNIT
V _(ESD) Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
	Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±250	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 500-V HBM is possible with the necessary precautions. Pins listed as ±2000 V may actually have higher performance.

- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 250-V CDM is possible with the necessary precautions. Pins listed as ±250 V may actually have higher performance.

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

	MIN	NOM	MAX	UNIT
V _S Total supply voltage	2.7		5.5	V
T _A Ambient temperature	–40	25	125	°C

7.4 Thermal Information: OPA355

THERMAL METRIC ⁽¹⁾		OPA355		UNIT
		D (SOIC)	DBV (SOT-23)	
		8 PINS	6 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	136.3	166.1	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	76.7	104.3	°C/W
R _{θJB}	Junction-to-board thermal resistance	79.8	38.7	°C/W
ψ _{JT}	Junction-to-top characterization parameter	26.3	23.4	°C/W
ψ _{JB}	Junction-to-board characterization parameter	79	38.5	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	—	—	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

7.5 Thermal Information: OPA2355

THERMAL METRIC ⁽¹⁾		OPA2355	UNIT
		DGS (VSSOP)	
		10 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	171.1	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	58	°C/W
R _{θJB}	Junction-to-board thermal resistance	92.9	°C/W
ψ _{JT}	Junction-to-top characterization parameter	6.7	°C/W
ψ _{JB}	Junction-to-board characterization parameter	91.2	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	—	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

7.6 Thermal Information: OPA3355

THERMAL METRIC ⁽¹⁾		OPA3355		UNIT
		D (SOIC)	PW (TSSOP)	
		14 PINS	14 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	85.3	113.3	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	41.4	38	°C/W
R _{θJB}	Junction-to-board thermal resistance	41.5	58.1	°C/W
ψ _{JT}	Junction-to-top characterization parameter	8.3	2.8	°C/W
ψ _{JB}	Junction-to-board characterization parameter	41.2	57.3	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	—	—	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

7.7 Electrical Characteristics: $V_S = 2.7\text{ V}$ to 5.5 V (Single-Supply)

at $T_A = 25^\circ\text{C}$, $R_F = 604\ \Omega$, $R_L = 150\ \Omega$, and connected to $V_S / 2$, (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
OFFSET VOLTAGE						
V _{OS}	Input offset voltage	V _S = 5 V		±2	±9	mV
		T _A = −40°C to +125°C			±15	
dV _{OS} /dT	Input offset voltage vs temperature	T _A = −40°C to +125°C		±7		μV/°C
	Input offset voltage vs power supply			±80	±350	μV/V
INPUT BIAS CURRENT						
I _B	Input bias current			3	±50	pA
I _{OS}	Input offset current			±1	±50	pA
NOISE						
	Input noise voltage density	f = 1 MHz		5.8		nV/√Hz
	Current noise density	f = 1 MHz		50		fA/√Hz
INPUT VOLTAGE RANGE						
V _{CM}	Common-mode voltage range		(V−) − 0.1	(V+) − 1.5		V
CMRR	Common-mode rejection ratio	V _S = 5.5 V, −0.1 V < V _{CM} < 4 V	66	80		dB
		V _S = 5.5 V, −0.1 V < V _{CM} < 4 V T _A = −40°C to +125°C	66			dB
INPUT IMPEDANCE						
	Differential			10 ¹³ 1.5		Ω pF
	Common-mode			10 ¹³ 1.5		Ω pF
OPEN-LOOP GAIN						
	Open-loop gain	V _S = 5 V, 0.3 V < V _O < 4.7 V	84	92		dB
		OPA355: V _S = 5 V, 0.3 V < V _O < 4.7 V T _A = −40°C to +125°C	80			dB
		OPA2355, OPA3355: V _S = 5 V, 0.4 V < V _O < 4.6 V	80			dB
FREQUENCY RESPONSE						
f _{−3dB}	Small-signal bandwidth	G = 1, V _O = 100 mVp-p, R _F = 0 Ω		450		MHz
		G = 2, V _O = 100 mVp-p, R _L = 50 Ω		100		MHz
		G = 2, V _O = 100 mVp-p, R _L = 150 Ω		170		MHz
		G = 2, V _O = 100 mVp-p, R _L = 1 kΩ		200		MHz
GBW	Gain-bandwidth product	G = 10, R _L = 1 kΩ		200		MHz
f _{0.1dB}	Bandwidth for 0.1-dB gain flatness	G = 2, V _O = 100 mVp-p, R _F = 560 Ω		75		MHz
SR	Slew rate	V _S = 5 V, G = 2, 4-V output step		300 / −360		V/μs
	Rise and fall time	G = 2, V _O = 200 Vp-p, 10% to 90%		2.4		ns
		G = 2, V _O = 2 Vp-p, 10% to 90%		8		ns
	Settling time	0.1%: V _S = 5 V, G = 2, 2-V output step		30		ns
		0.01%: V _S = 5 V, G = 2, 2-V output step		120		ns
	Overload recovery time	V _{IN} × gain = V _S		8		ns
HARMONIC DISTORTION						
	Second harmonic	G = 2, f = 1 MHz, V _O = 2 Vp-p, R _L = 200 Ω		−81		dBc
	Third harmonic	G = 2, f = 1 MHz, V _O = 2 Vp-p, R _L = 200 Ω		−93		dBc
	Differential gain error	NTSC, R _L = 150 Ω		0.02		%

Electrical Characteristics: $V_S = 2.7\text{ V}$ to 5.5 V (Single-Supply) (continued)

at $T_A = 25^\circ\text{C}$, $R_F = 604\ \Omega$, $R_L = 150\ \Omega$, and connected to $V_S / 2$, (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Differential phase error		NTSC, R _L = 150 Ω		0.05		°
Channel-to-channel crosstalk		OPA2355: f = 5 MHz		−90		dB
		OPA3335: f = 5 MHz		−70		dB
OUTPUT						
Voltage output swing from rail		V _S = 5 V, R _L = 150 Ω, A _{OL} > 84 dB		0.2	0.3	V
		V _S = 5 V, R _L = 1 kΩ		0.1		V
I _O	Continuous output current ⁽¹⁾			±60		mA
	Peak output current ⁽¹⁾	V _S = 5 V		±100		mA
		V _S = 3 V		±80		mA
	Closed-loop output impedance ⁽¹⁾	f < 100 kHz		0.02		Ω
POWER SUPPLY						
V _S	Specified voltage range		2.7		5.5	V
	Operating voltage range		2.5 to 5.5			V
I _Q	Quiescent current (per amplifier)	V _S = 5 V, enabled, I _O = 0		8.3	11	mA
		T _A = −40°C to +125°C			14	mA
SHUTDOWN						
	Disabled (logic-LOW threshold)				0.8	V
	Enabled (logic-HIGH threshold)		2			V
	Enable time			100		ns
	Disable time			30		ns
	Shutdown current (per amplifier)	V _S = 5 V, disabled		3.4	6	μA
THERMAL SHUTDOWN						
Junction temperature		Shutdown		160		°C
		Reset from shutdown		140		
TEMPERATURE RANGE						
	Specified range		−40		125	°C
	Operating range		−55		150	°C
	Storage range		−65		150	°C
θ _{JA}	Thermal resistance	SOT-23-6, VSSOP-10		150		°C/W
		SO-8		125		°C/W
		SO-14, TSSOP-14		100		°C/W

(1) See [Output Voltage Swing vs Output Current](#).

7.8 Typical Characteristics

$T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $G = 2$, $R_F = 604\ \Omega$, and $R_L = 150\ \Omega$ connected to $V_S / 2$, (unless otherwise noted)

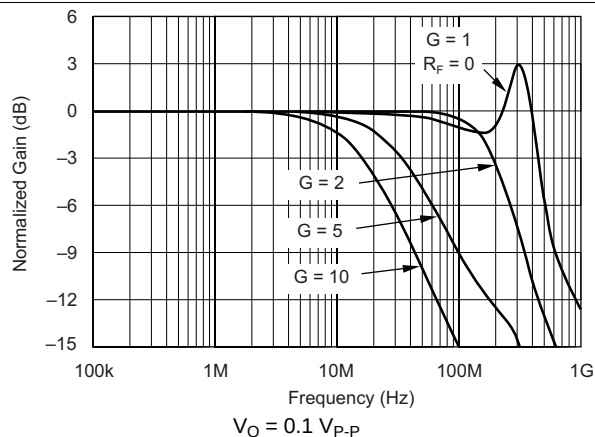


Figure 1. Noninverting Small-Signal Frequency Response

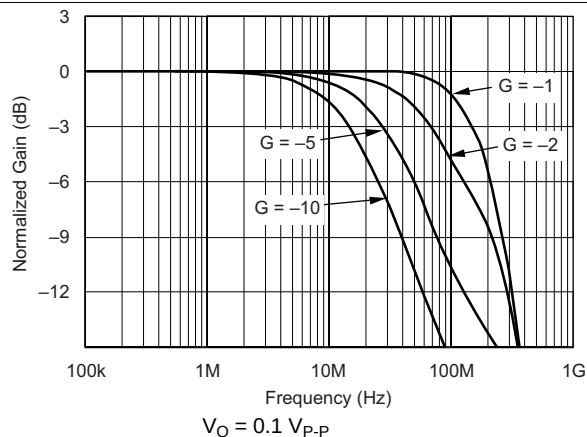


Figure 2. Inverting Small-Signal Frequency Response

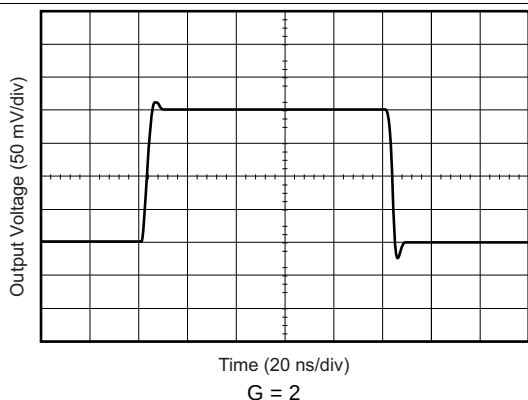


Figure 3. Noninverting Small-Signal Step Response

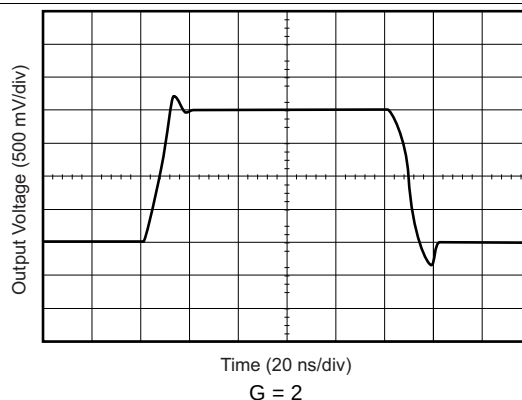


Figure 4. Noninverting Large-Signal Step Response

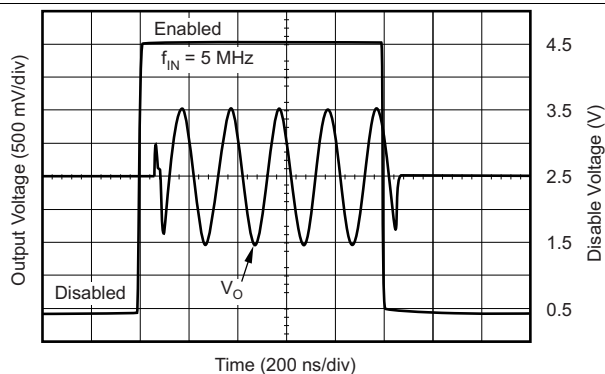


Figure 5. Large-Signal Disable and Enable Response

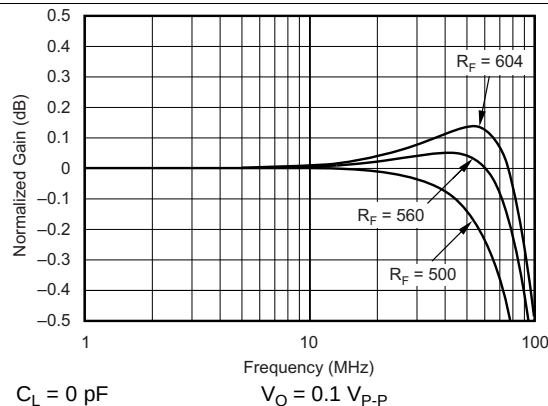


Figure 6. 0.1-dB Gain Flatness for Various R_F Values

Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $G = 2$, $R_F = 604\ \Omega$, and $R_L = 150\ \Omega$ connected to $V_S / 2$, (unless otherwise noted)

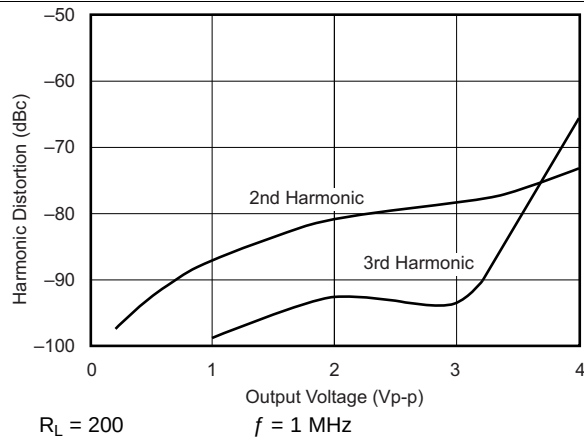


Figure 7. Harmonic Distortion vs Output Voltage

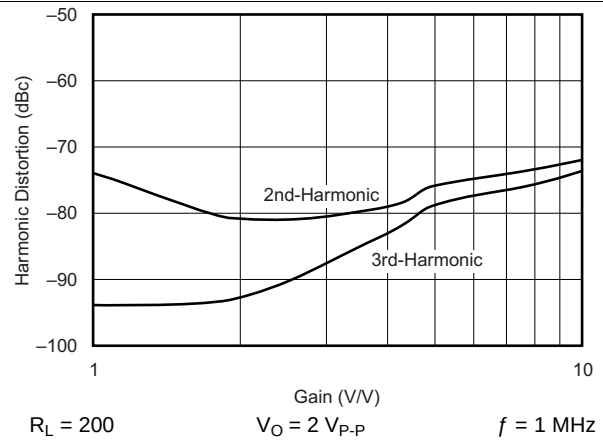


Figure 8. Harmonic Distortion vs Noninverting Gain

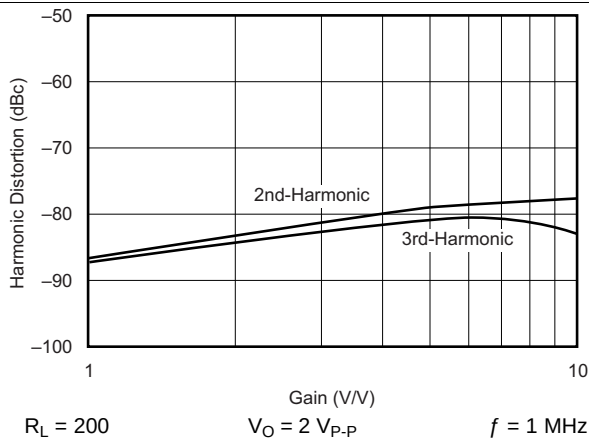


Figure 9. Harmonic Distortion vs Inverting Gain

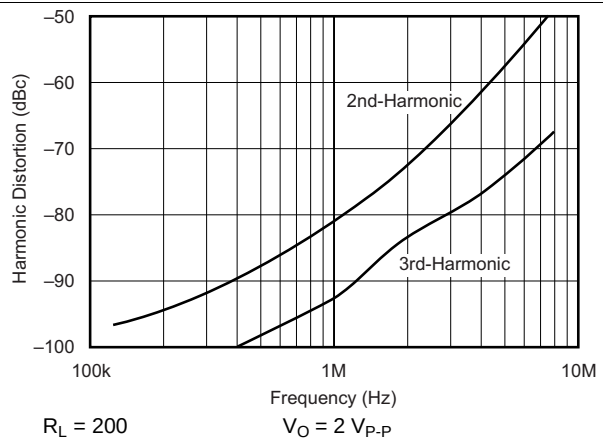


Figure 10. Harmonic Distortion vs Frequency

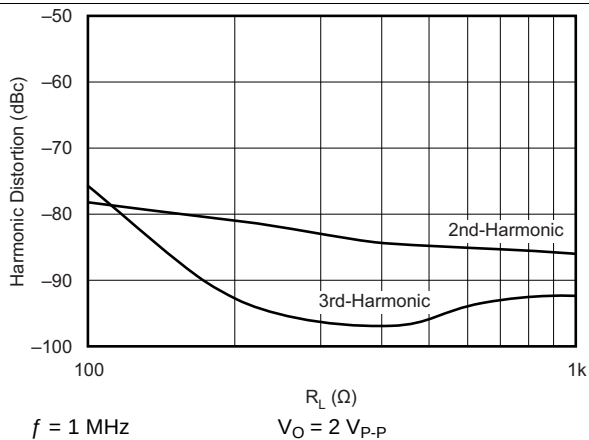


Figure 11. Harmonic Distortion vs Load Resistance

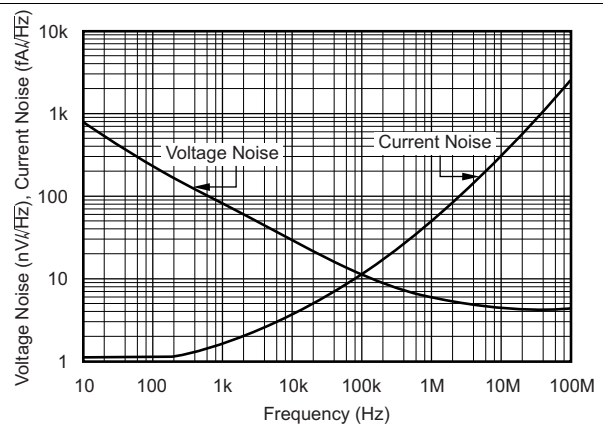


Figure 12. Input Voltage and Current Noise Spectral Density vs Frequency

Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $G = 2$, $R_F = 604\ \Omega$, and $R_L = 150\ \Omega$ connected to $V_S / 2$, (unless otherwise noted)

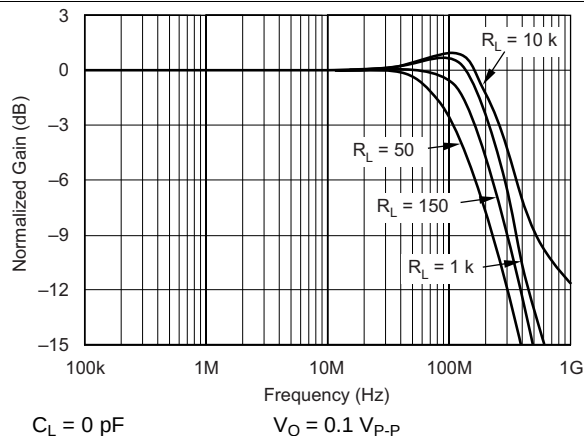


Figure 13. Frequency Response for Various R_L Values

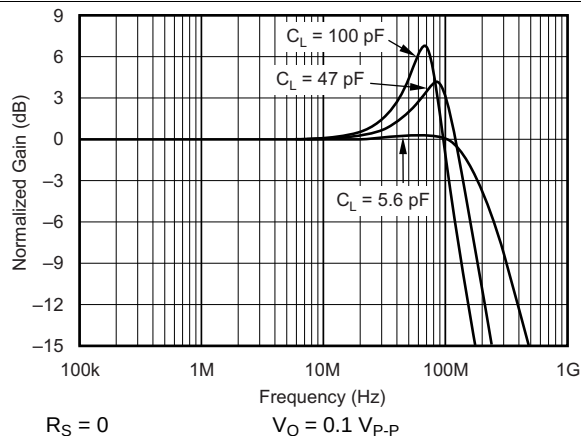


Figure 14. Frequency Response for Various C_L Values

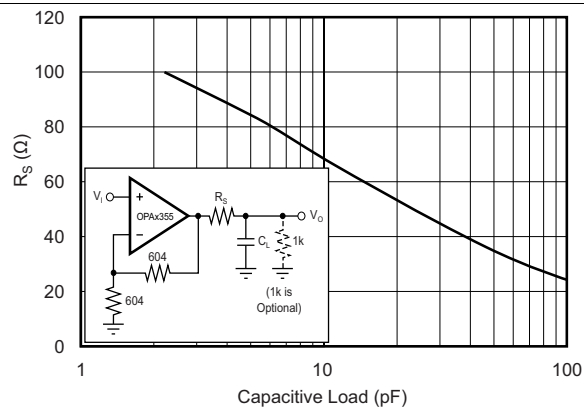


Figure 15. Recommended R_S Values vs Capacitive Load

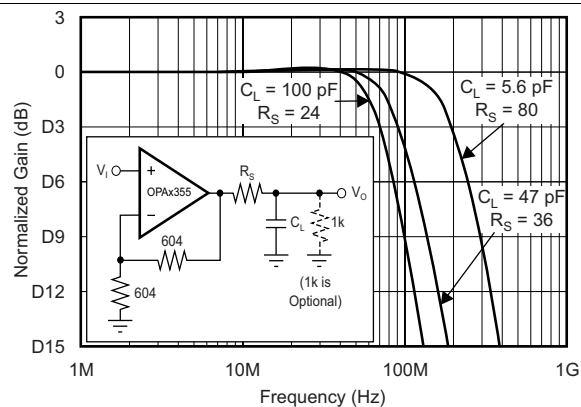


Figure 16. Frequency Response vs Capacitive Load

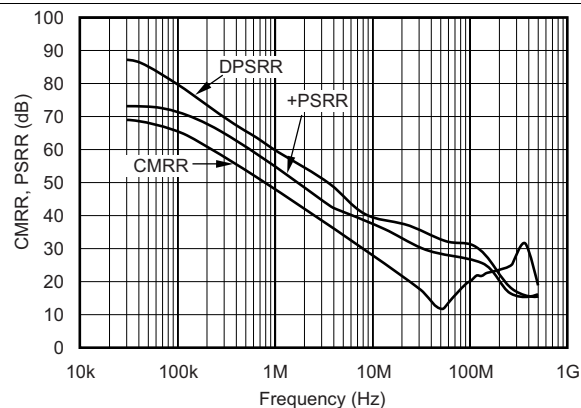


Figure 17. Common-Mode Rejection Ratio and Power-Supply Rejection Ratio vs Frequency

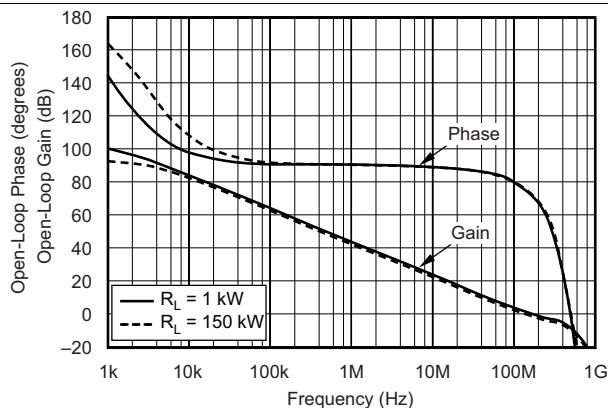


Figure 18. Open-Loop Gain and Phase

Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $G = 2$, $R_F = 604\ \Omega$, and $R_L = 150\ \Omega$ connected to $V_S / 2$, (unless otherwise noted)

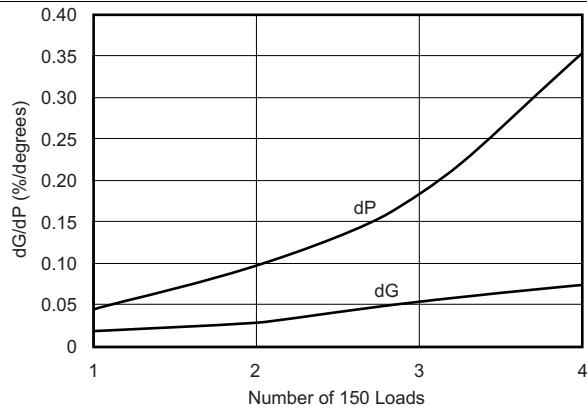


Figure 19. Composite Video Differential Gain and Phase

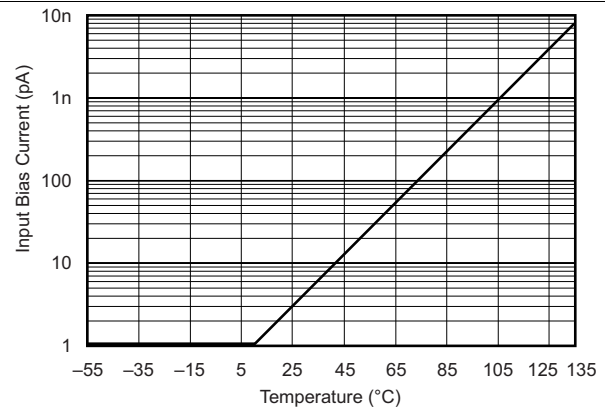


Figure 20. Input Bias Current vs Temperature

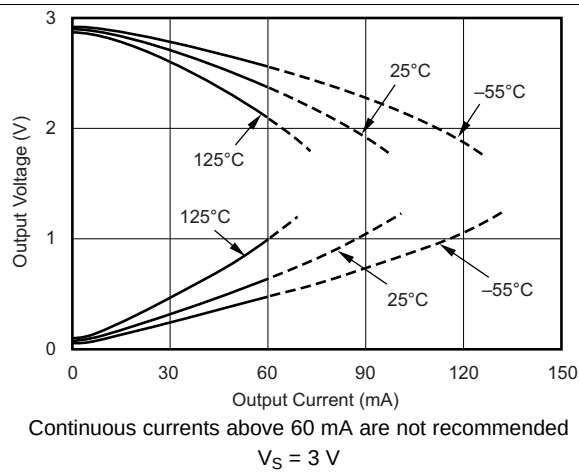


Figure 21. Output Voltage Swing vs Output Current

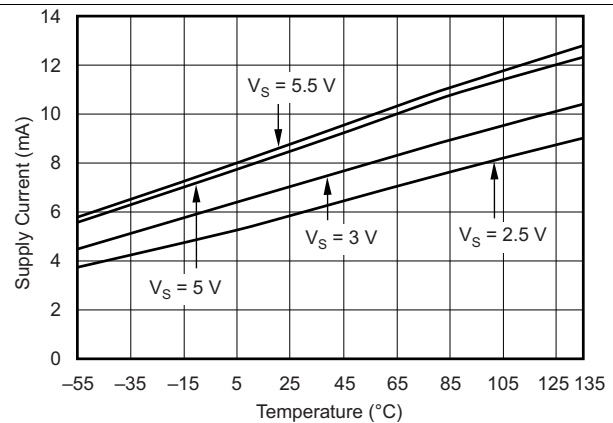


Figure 22. Supply Current vs Temperature

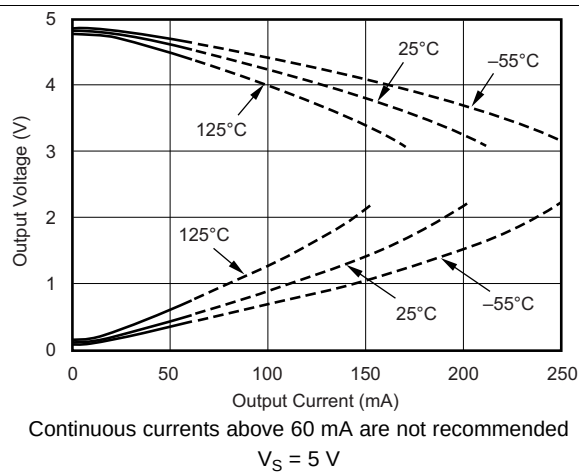


Figure 23. Output Voltage Swing vs Output Current

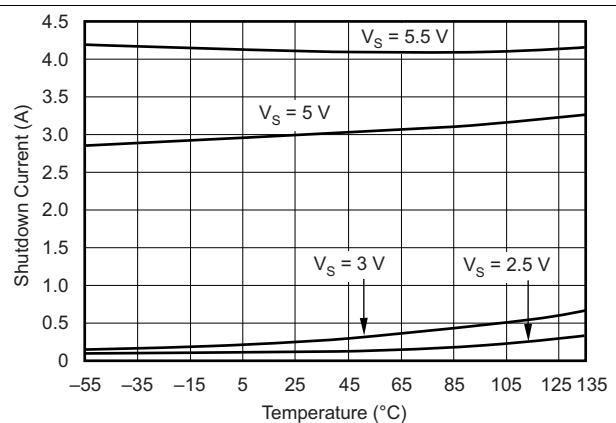


Figure 24. Shutdown Current vs Temperature

Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$, $V_S = 5\text{ V}$, $G = 2$, $R_F = 604\ \Omega$, and $R_L = 150\ \Omega$ connected to $V_S / 2$, (unless otherwise noted)

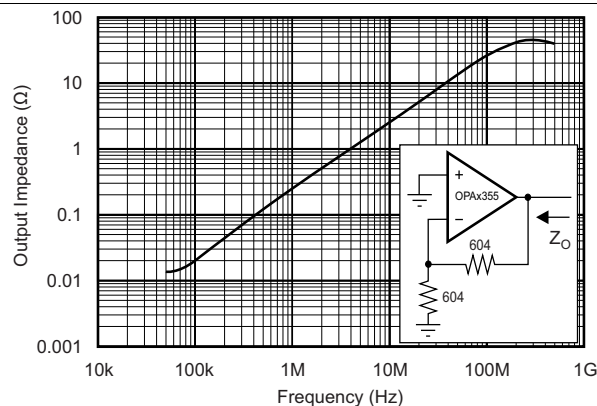
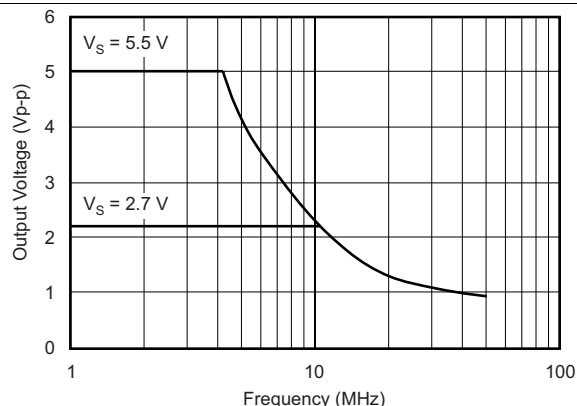


Figure 25. Closed-Loop Output Impedance vs Frequency



Maximum output voltage without slew-rate induced distortion

Figure 26. Maximum Output Voltage vs Frequency

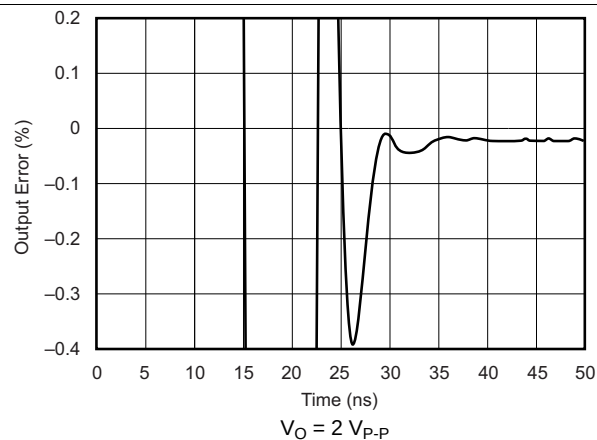


Figure 27. Output Settling Time to 0.1%

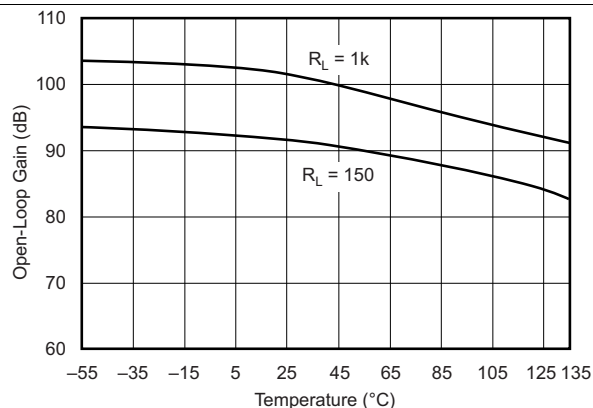


Figure 28. Open-Loop Gain vs Temperature

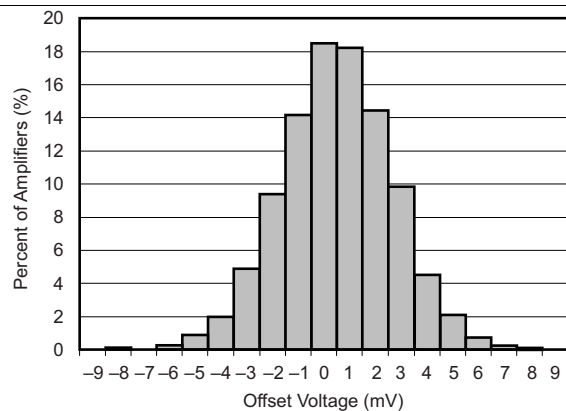


Figure 29. Offset Voltage Production Distribution

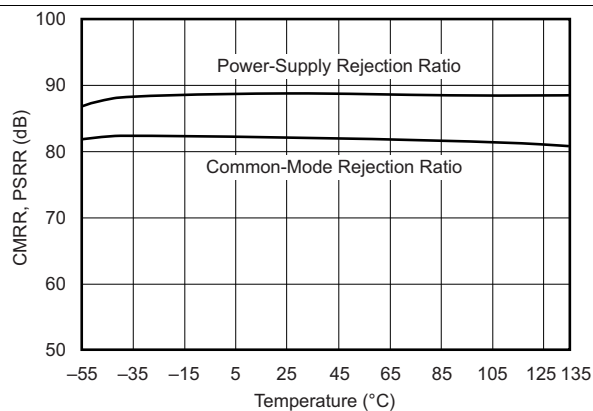


Figure 30. Common-Mode Rejection Ratio and Power-Supply Rejection Ratio vs Temperature

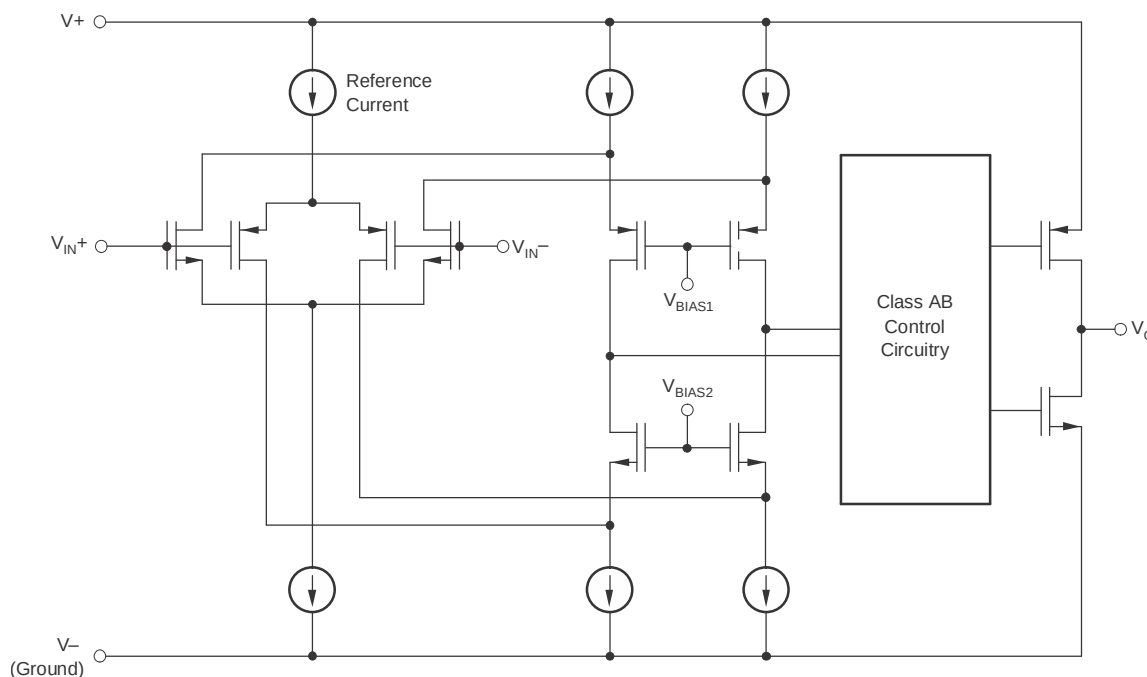
8 Detailed Description

8.1 Overview

The OPA355 series is a CMOS, high-speed, voltage-feedback, operational amplifier designed for video and other general-purpose applications. The series is available as a single, dual, or triple op amp. The family features a 200-MHz gain bandwidth and 360 V/ μ s slew rate, but the series is unity-gain stable and can operate as a 1 V/V voltage follower.

The input common-mode range includes ground, allowing the OPAx355 family to be used in virtually any single-supply application up to a supply voltage of 5.5 V.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Operating Voltage

The OPAx355 family is specified over a power-supply range of 2.7 V to 5.5 V (± 1.35 to ± 2.75 V). However, the supply voltage ranges from 2.5 to 5.5 V (± 1.25 to ± 2.75 V). Supply voltages higher than 7.5 V (absolute maximum) can permanently damage the amplifier.

Parameters that vary significantly over supply voltage or temperature are shown in the [Typical Characteristics](#) section of this data sheet.

8.3.2 Enable Function

The OPAx355 series is enabled by applying a TTL high-voltage level to the enable pin. Conversely, a TTL low-voltage level disables the amplifier, which reduces the supply current from 8.3 mA to 3.4 μ A per amplifier. This pin voltage is referenced to a single-supply ground. When using a split-supply, such as ± 2.5 V, the enable and disable voltage levels are referenced to V-. For portable battery-operated applications, this feature greatly reduces the average current and as a result, extends battery life.

The enable input is modeled as a CMOS input gate with a 100-k Ω pullup resistor to V+. The enable pin assumes a logic high and the amplifier turns on if the enable pin is left open.

Feature Description (continued)

The enable time is 100 ns and the disable time is 30 ns, which allows the OPAx355 series to operate as a *gated* amplifier, or to have the output multiplexed onto a common output bus. When disabled, the output assumes a high-impedance state.

8.3.3 Output Drive

The output stage supplies a high short-circuit current (typically over 200 mA). Therefore, an on-chip thermal shutdown circuit is provided to protect the OPAx355 series from dangerously-high junction temperatures. At 160°C, the protection circuit shuts down the amplifier. Normal operation resumes when the junction temperature cools to below 140°C.

NOTE

Running a continuous DC current in excess of ± 60 mA is not recommended. See the *Output Voltage Swing vs Output Current* graphs ([Figure 21](#) and [Figure 22](#)) in the [Typical Characteristics](#) section.

8.4 Device Functional Modes

The OPAx355 family is powered on when the supply is connected. The series operates as a single supply operational amplifier or dual supply amplifier depending on the application. The series is used with asymmetrical supplies as long as the differential voltage (V^- to V^+) is at least 1.8 V and no greater than 5.5 V (example: V^- set to -3.5 V and V^+ set to 1.5 V).

9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The OPAx355 series is a CMOS, high-speed, voltage-feedback, operational amplifier (op amp) designed for general-purpose applications.

The amplifiers feature a 200-MHz gain bandwidth and 300-V/ μ s slew rate, but the devices are unity-gain stable and operate as a 1-V/V voltage follower.

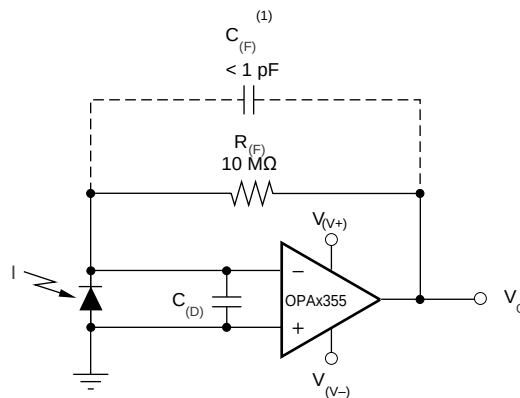
The input common-mode voltage range of the series includes ground, which allows the OPAx355 to be used in virtually any single-supply application up to a supply voltage of 5.5 V.

9.2 Typical Applications

9.2.1 Transimpedance Amplifier

Wide gain bandwidth, low input bias current, low input voltage, and current noise make the OPAx355 series a preferred wideband photodiode transimpedance amplifier family. Low voltage noise is important because photodiode capacitance causes the effective noise gain of the circuit to increase at high frequencies.

The key elements to a transimpedance design, as shown in Figure 31, are the expected diode capacitance ($C_{[D]}$), which must include the parasitic input common-mode and differential-mode input capacitance (4 pF + 5 pF), the desired transimpedance gain ($R_{[FB]}$), and the gain-bandwidth (GBW) for the OPAx355 family (20 MHz). With these three variables set, the feedback capacitor value ($C_{[FB]}$) controls the frequency response. $C_{[FB]}$ includes the stray capacitance of $R_{[FB]}$, which is 0.2 pF for a typical surface-mount resistor.



(1) $C_{(FB)}$ is optional to prevent gain peaking. $C_{(FB)}$ includes the stray capacitance of $R_{(FB)}$.

Figure 31. Dual-Supply Transimpedance Amplifier

9.2.1.1 Design Requirements

PARAMETER	VALUE
Supply voltage $V_{(V+)}$	2.5 V
Supply voltage $V_{(V-)}$	-2.5 V

9.2.1.2 Detailed Design Procedure

To achieve a maximally-flat, second-order Butterworth frequency response, set the feedback pole to:

$$\frac{1}{2 \times \pi \times R_{(FB)} \times C_{(FB)}} = \sqrt{\frac{GBW}{4 \times \pi \times R_{(FB)} \times C_{(D)}}} \quad (1)$$

Use Equation 2 to calculate the bandwidth.

$$f_{(-3 \text{ dB})} = \sqrt{\frac{GBW}{2 \times \pi \times R_{(FB)} \times C_{(D)}}} \quad (2)$$

For other transimpedance bandwidths, consider the high-speed CMOS OPA380 (90-MHz GBW), OPA354 (100-MHz GBW), OPA300 (180-MHz GBW), OPA355 (200-MHz GBW), or OPA656 and OPA657 (400-MHz GBW).

For single-supply applications, the +INx input is biased with a positive DC voltage to allow the output to reach true zero when the photodiode is not exposed to any light, and respond without the added delay that results from coming out of the negative rail. Figure 32 shows this configuration. This bias voltage appears across the photodiode, providing a reverse bias for faster operation.

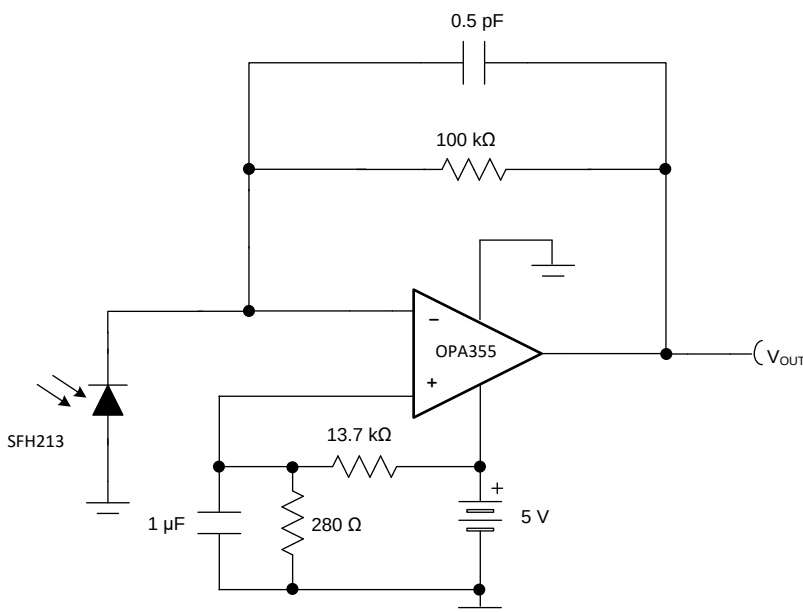


Figure 32. Single-Supply Transimpedance Amplifier

For additional information, see [Compensate Transimpedance Amplifiers Intuitively](#).

9.2.1.2.1 Optimizing The Transimpedance Circuit

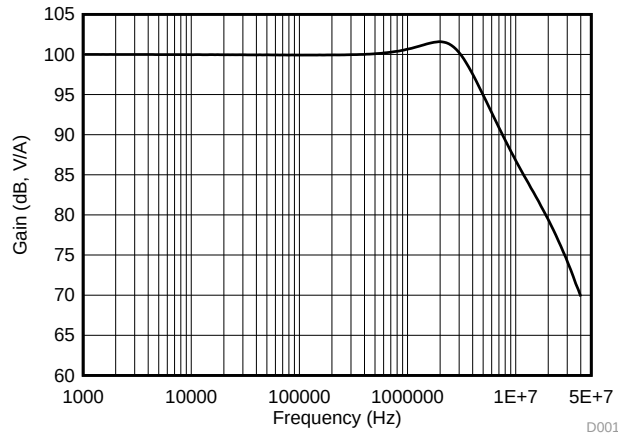
To achieve the best performance, select components according to the following guidelines:

1. For lowest noise, select $R_{(FB)}$ to create the total required gain. Using a lower value for $R_{(FB)}$ and adding gain after the transimpedance amplifier generally results in poorer noise performance. $R_{(FB)}$ produces noise that increases with the square root of $R_{(FB)}$, whereas the signal increases linearly. Therefore, signal-to-noise ratio improves when all the required gain is placed in the transimpedance stage.
2. Minimize photodiode capacitance and stray capacitance at the summing junction (inverting input). This capacitance causes the voltage noise of the op amp to amplify (increasing amplification at high frequencies). Using a low-noise voltage source to reverse-bias a photodiode can significantly reduce the capacitance. Smaller photodiodes have lower capacitance. Use optics to concentrate light on a small photodiode.
3. Noise increases with increased bandwidth. Only use the required circuit bandwidth. Use a capacitor across the $R_{(FB)}$ to limit bandwidth, even if a capacitor is not required for stability.
4. Circuit board leakage can degrade the performance of an otherwise well-designed amplifier. Clean the circuit board carefully. Control leakage by using a circuit board guard trace that encircles the summing junction and

drives at the same voltage.

For additional information, see [Noise Analysis of FET Transimpedance Amplifiers](#) and [Noise Analysis for High-Speed Op Amps](#)).

9.2.1.3 Application Curve



–3 dB bandwidth is 4.56 MHz

Figure 33. AC Transfer Function

9.2.2 High-Impedance Sensor Interface

Many sensors have high source impedances that may range up to 10 MΩ, or even higher. The output signal of sensors often must be amplified or otherwise conditioned by an amplifier. The input bias current of this amplifier loads the sensor output and causes a voltage drop across the source resistance, [Figure 34](#) shows $(V_{(+INx)} = V_S - I_{(BIAS)} \times R_{(S)})$. The last term $(I_{(BIAS)} \times R_{(S)})$ shows the voltage drop across $R_{(S)}$. To prevent errors introduced to the system as a result of this voltage, use an op amp with low input bias current with high-impedance sensors. This low current keeps the $I_{(BIAS)} \times R_{(S)}$ error contribution less than the input voltage noise of the amplifier so that input voltage noise is not the dominant noise factor. The OPAx355 op amps feature low input bias current (typically 200 fA), and as a result, a preferred choice for these applications.

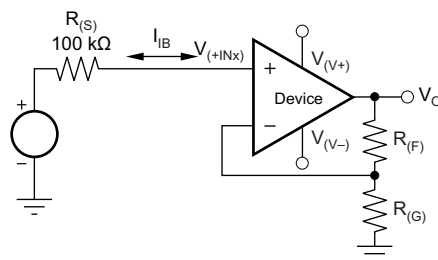
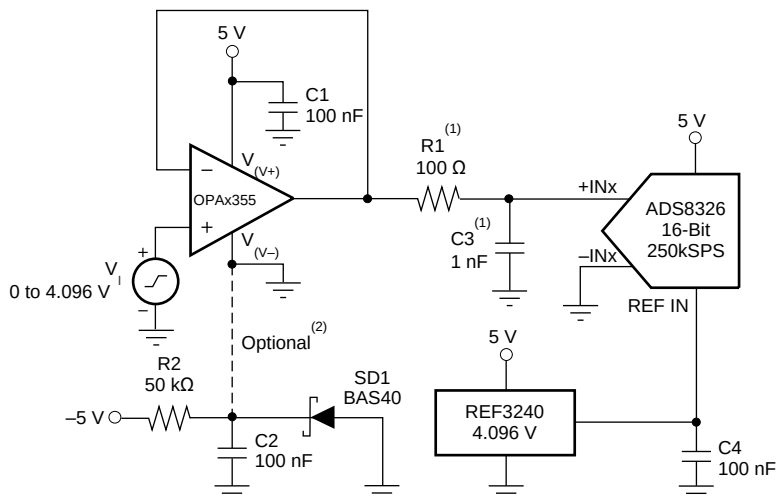


Figure 34. Noise as a Result of $I_{(BIAS)}$

9.2.3 Driving ADCs

The OPAx355 op amps are designed to drive sampling analog-to-digital converters (ADCs) with sampling speeds up to 1 MSPS. The zero-crossover distortion input stage topology allows the OPAx355 series to drive ADCs without degrading differential linearity and THD.

The OPAx355 series buffers the ADC switched input capacitance and resulting charge injection while providing signal gain. [Figure 35](#) shows the OPAx355 series configured to drive the [ADS8326](#).



(1) Suggested value; may require adjustment based on specific application.

(2) Single-supply applications lose a small number of ADC codes near ground as a result of op amp output swing limitation. If a negative power supply is available, this simple circuit creates a -0.3V supply to allow output swing to true ground potential.

Figure 35. Driving the ADS8326

9.2.4 Active Filter

The OPAx355 series is designed for active filter applications that require a wide bandwidth, fast slew rate, low-noise, single-supply operational amplifier. [Figure 36](#) shows a 500 kHz, second-order, low-pass filter using the multiple-feedback (MFB) topology. The components are selected to provide a maximally-flat Butterworth response. Beyond the cutoff frequency, roll-off is -40 dB/dec . The Butterworth response is preferred for applications requiring predictable gain characteristics, such as the anti-aliasing filter used in front of an ADC.

One point to observe when considering the MFB filter is that the output is inverted, relative to the input. If this inversion is not required, or not desired, a noninverting output can be achieved through one of the following options:

1. Adding an inverting amplifier
2. Adding an additional second-order MFB stage
3. Using a noninverting filter topology, such as the Sallen-Key (see [Figure 37](#)).

MFB and Sallen-Key, low-pass and high-pass filter synthesis is quickly accomplished using TI's [FilterPro™](#) program. This software is available as a free download at www.ti.com.

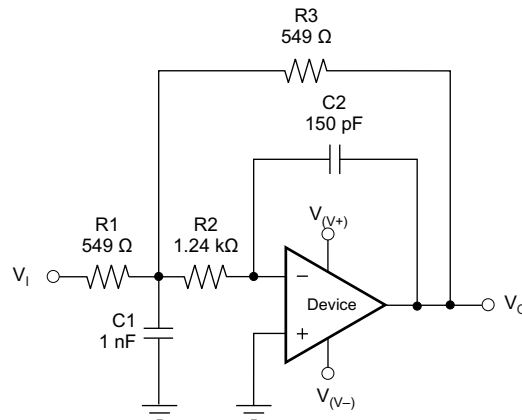


Figure 36. Second-Order Butterworth 500-kHz Low-Pass Filter

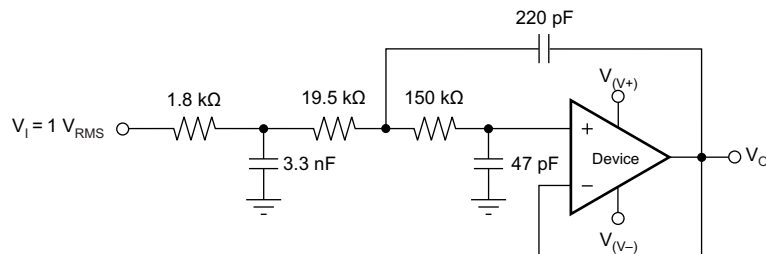


Figure 37. OPAx355 Configured as a Three-Pole, 20-kHz, Sallen-Key Filter

9.3 Video

The OPAx355 output stage is capable of driving a standard back-terminated 75-Ω video cable. By back-terminating a transmission line, the line does not exhibit a capacitive load to the driver. A properly back-terminated 75-Ω cable does not appear as capacitance; the cable presents only a 150-Ω resistive load to the OPAx355 output.

The OPAx355 can be used as an amplifier for RGB graphic signals, which have a voltage of zero at the video black level by offsetting and AC-coupling the signal, as shown in [Figure 38](#).

Video (continued)

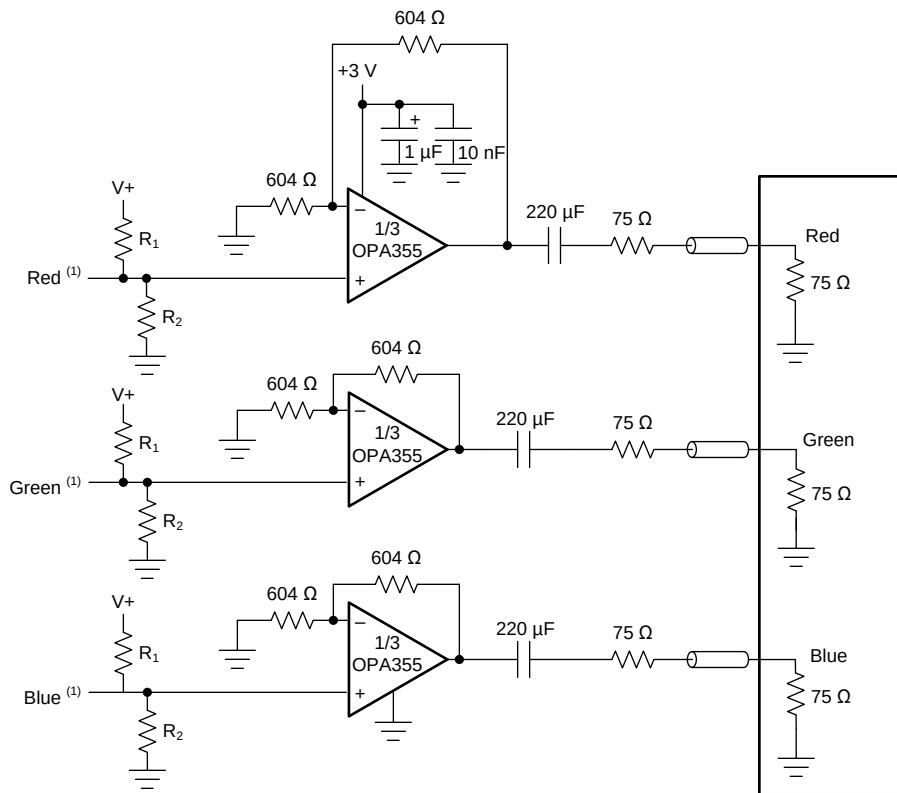


Figure 38. RGB Cable Driver

9.4 Wideband Video Multiplexing

One common application for video speed amplifiers which include an enable pin is to wire multiple amplifier outputs together, then select which one of several possible video inputs to source onto a single line. This simple wired-OR video multiplexer can be easily implemented using the OPA357; see [Figure 39](#).

Wideband Video Multiplexing (continued)

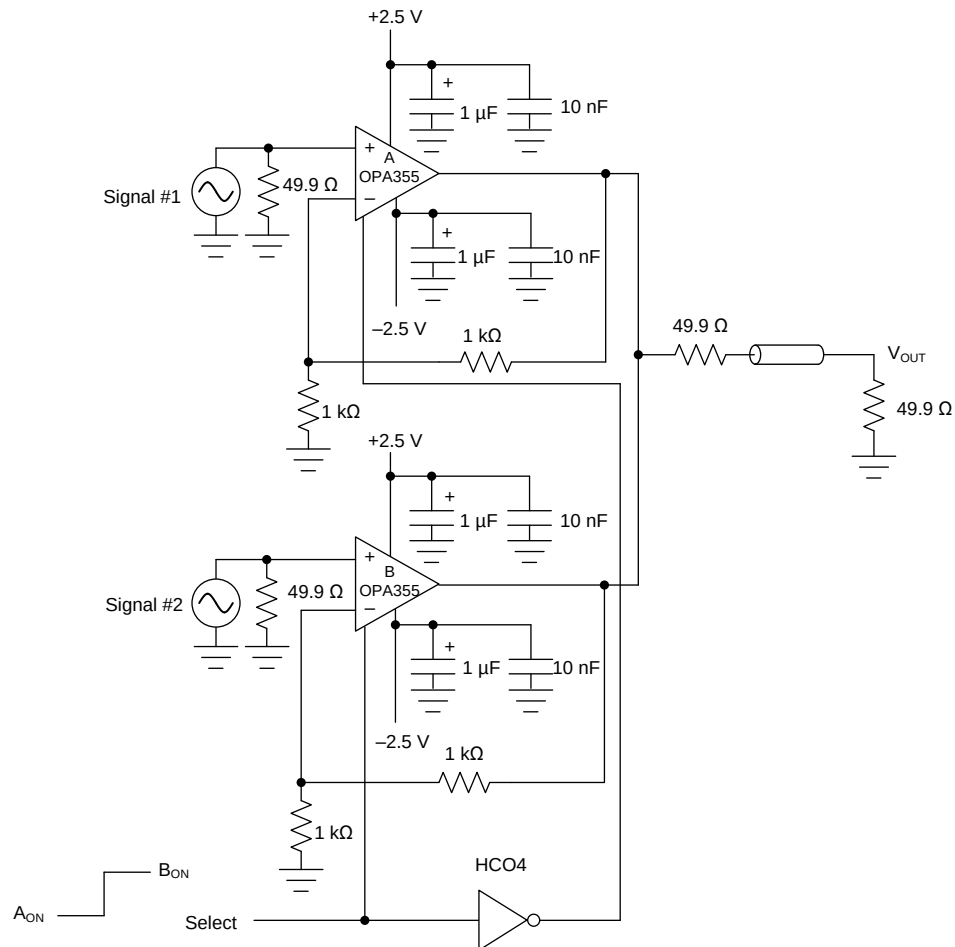


Figure 39. Multiplexed Output

10 Power Supply Recommendations

The OPAx355 is specified for operation from 2.7 to 5.5 V (± 1.35 to ± 2.75 V); many specifications apply from -40°C to $+125^{\circ}\text{C}$. Parameters that can exhibit significant variance with regard to operating voltage or temperature are shown in the [Typical Characteristics](#) section.

Place 0.1- μF bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high-impedance power supplies. For more detailed information on bypass capacitor placement, see the [Layout Guidelines](#) section.

Power dissipation depends on power-supply voltage, signal and load conditions. With DC signals, power dissipation is equal to the product of output current times the voltage across the conducting output transistor, $V_S - V_O$. Minimize power dissipation by using the lowest possible power-supply voltage required to ensure the required output voltage swing.

For resistive loads, the maximum power dissipation occurs at a DC output voltage of one-half the power-supply voltage. Dissipation with AC signals is lower. *Power Amplifier Stress and Power Handling Limitations* explains how to calculate or measure power dissipation with unusual signals and loads, and is available on [www.ti.com](#).

Any tendency to activate the thermal protection circuit indicates excessive power dissipation or an inadequate heat sink. For reliable operation, limit junction temperature to 150°C maximum. To estimate the margin of safety in a complete design, increase the ambient temperature to trigger the thermal protection at 160°C . The thermal protection must trigger more than 35°C above the maximum expected ambient condition of the application.

11 Layout

11.1 Layout Guidelines

Good high-frequency printed-circuit board (PCB) layout techniques must be used for the OPAx355 amplifiers. Generous use of ground planes, short direct-signal traces, and a preferred bypass capacitor located at the V+ pin ensures clean and stable operation. Large areas of copper help dissipate heat generated within the amplifiers in normal operation.

Sockets are not recommended for use with any high-speed amplifier.

A 10-nF ceramic bypass capacitor is the minimum recommended value; adding a 1- μF or larger tantalum capacitor in parallel is beneficial when driving a low-resistance load. Providing adequate bypass capacitance is essential to achieving very low harmonic and intermodulation distortion.

11.2 Layout Example

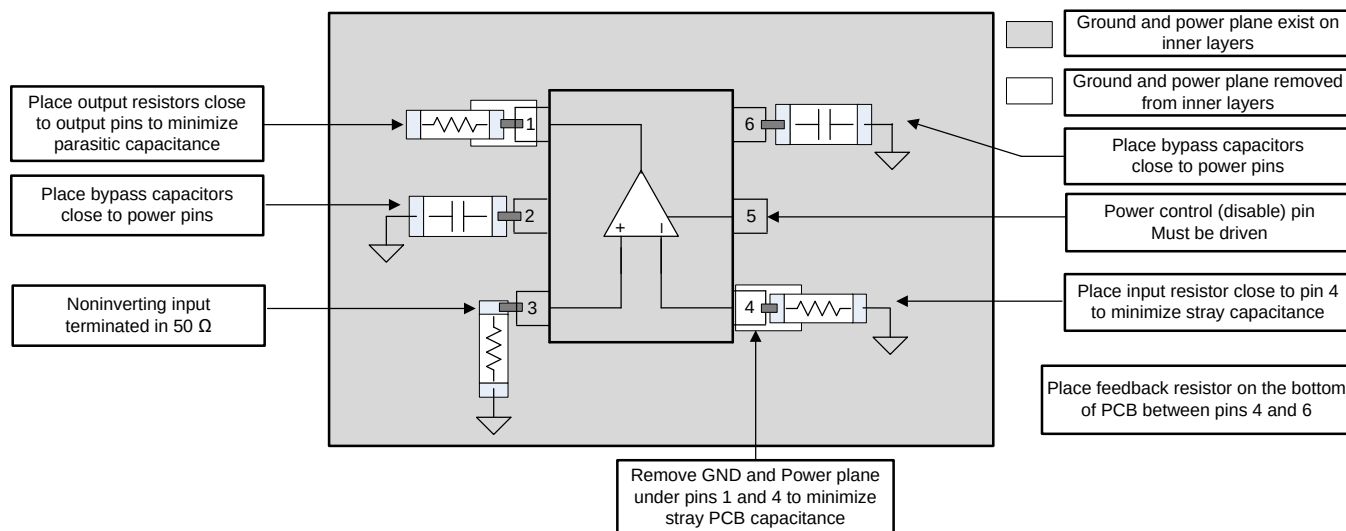


Figure 40. Layout Example

12 Device and Documentation Support

12.1 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to order now.

Table 1. Related Links

PARTS	PRODUCT FOLDER	ORDER NOW	TECHNICAL DOCUMENTS	TOOLS & SOFTWARE	SUPPORT & COMMUNITY
OPA355	Click here	Click here	Click here	Click here	Click here
OPA2355	Click here	Click here	Click here	Click here	Click here
OPA3355	Click here	Click here	Click here	Click here	Click here

12.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

12.3 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

12.4 Trademarks

E2E is a trademark of Texas Instruments.

FilterPro is a trademark of Texas Instruments Incorporated.

All other trademarks are the property of their respective owners.

12.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

12.6 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
OPA2355DGSA/250	ACTIVE	VSSOP	DGS	10	250	RoHS & Green	NIPDAUAG	Level-2-260C-1 YEAR	-40 to 125	D55	Samples
OPA2355DGSA/250G4	ACTIVE	VSSOP	DGS	10	250	RoHS & Green	NIPDAUAG	Level-2-260C-1 YEAR	-40 to 125	D55	Samples
OPA3355EA/250	ACTIVE	TSSOP	PW	14	250	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA 3355EA	Samples
OPA3355EA/2K5	ACTIVE	TSSOP	PW	14	2500	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA 3355EA	Samples
OPA3355UA	ACTIVE	SOIC	D	14	50	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA3355UA	Samples
OPA355NA/250	ACTIVE	SOT-23	DBV	6	250	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	C55	Samples
OPA355NA/250G4	ACTIVE	SOT-23	DBV	6	250	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	C55	Samples
OPA355NA/3K	ACTIVE	SOT-23	DBV	6	3000	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	C55	Samples
OPA355NA/3KG4	ACTIVE	SOT-23	DBV	6	3000	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	C55	Samples
OPA355UA	ACTIVE	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA 355UA	Samples
OPA355UA/2K5	ACTIVE	SOIC	D	8	2500	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA 355UA	Samples
OPA355UAG4	ACTIVE	SOIC	D	8	75	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	OPA 355UA	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

- ⁽³⁾ MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.
- ⁽⁴⁾ There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.
- ⁽⁵⁾ Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.
- ⁽⁶⁾ Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

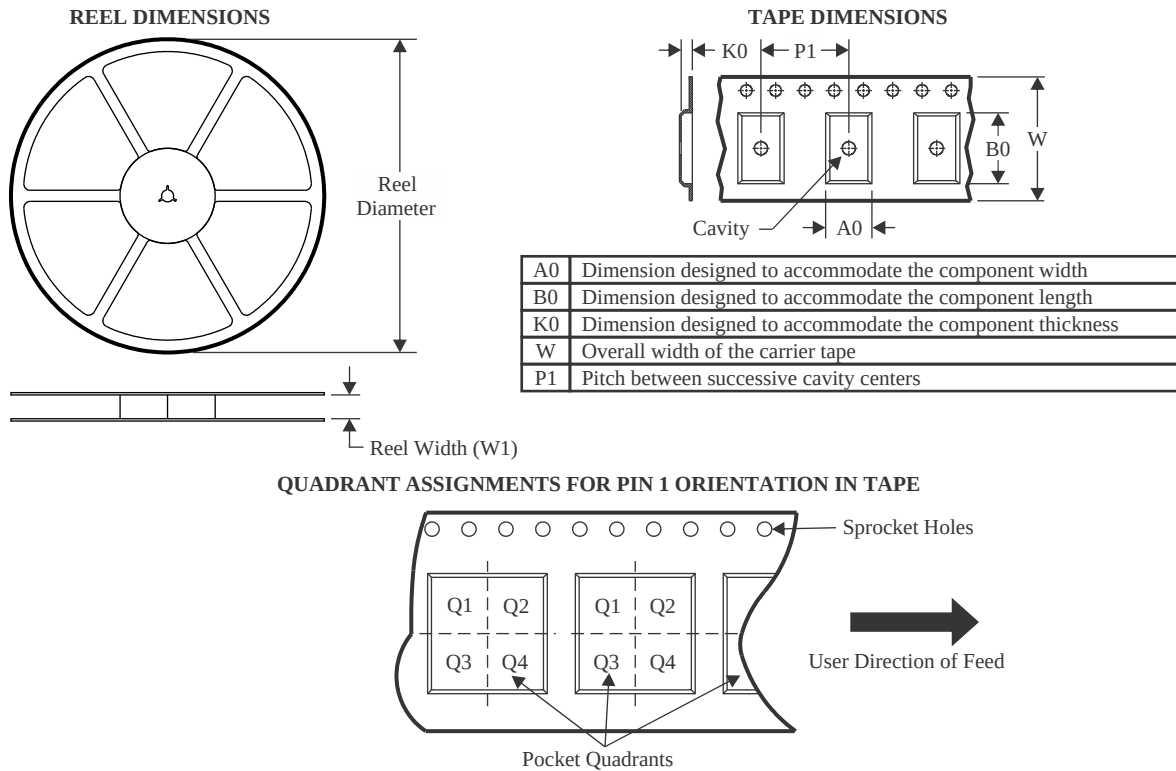
OTHER QUALIFIED VERSIONS OF OPA355 :

- Automotive : [OPA355-Q1](#)

NOTE: Qualified Version Definitions:

- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

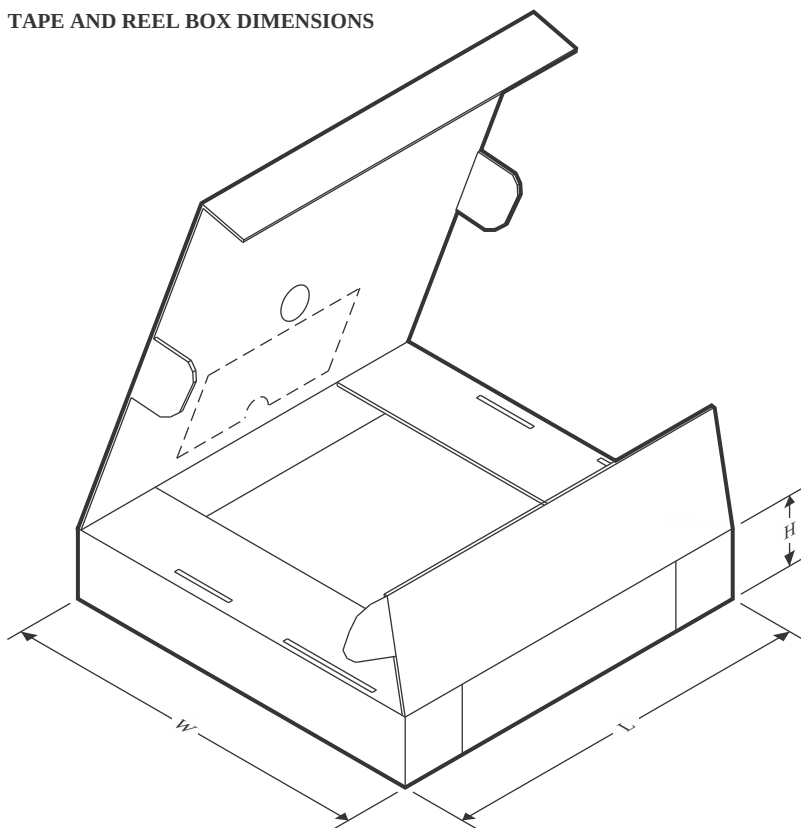
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
OPA2355DGSA/250	VSSOP	DGS	10	250	180.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
OPA3355EA/250	TSSOP	PW	14	250	180.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
OPA3355EA/2K5	TSSOP	PW	14	2500	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
OPA355NA/250	SOT-23	DBV	6	250	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
OPA355NA/3K	SOT-23	DBV	6	3000	178.0	9.0	3.23	3.17	1.37	4.0	8.0	Q3
OPA355UA/2K5	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1

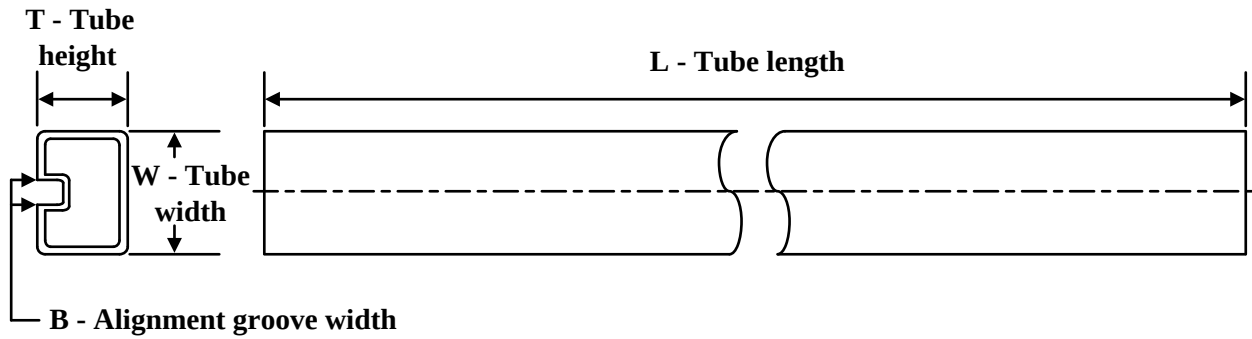
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
OPA2355DGSA/250	VSSOP	DGS	10	250	210.0	185.0	35.0
OPA3355EA/250	TSSOP	PW	14	250	210.0	185.0	35.0
OPA3355EA/2K5	TSSOP	PW	14	2500	356.0	356.0	35.0
OPA355NA/250	SOT-23	DBV	6	250	445.0	220.0	345.0
OPA355NA/3K	SOT-23	DBV	6	3000	445.0	220.0	345.0
OPA355UA/2K5	SOIC	D	8	2500	356.0	356.0	35.0

TUBE



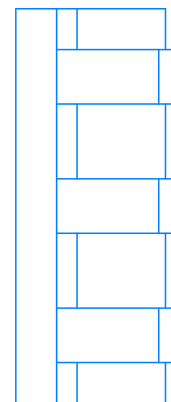
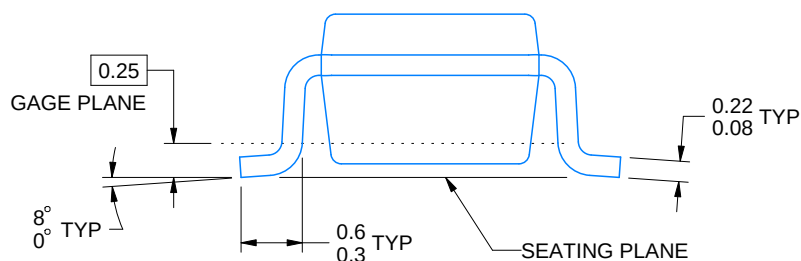
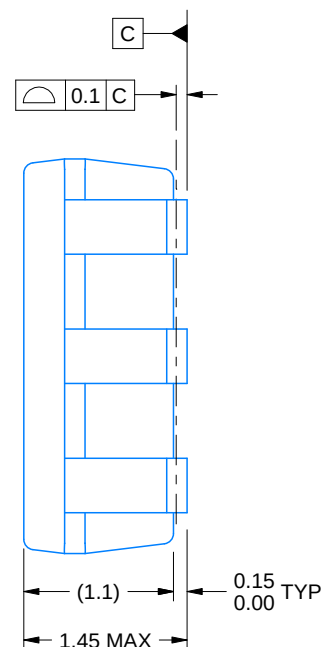
*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
OPA3355UA	D	SOIC	14	50	506.6	8	3940	4.32
OPA355UA	D	SOIC	8	75	506.6	8	3940	4.32
OPA355UAG4	D	SOIC	8	75	506.6	8	3940	4.32



SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



ALTERNATIVE PACKAGE SINGULATION VIEW

4214840/E 02/2024

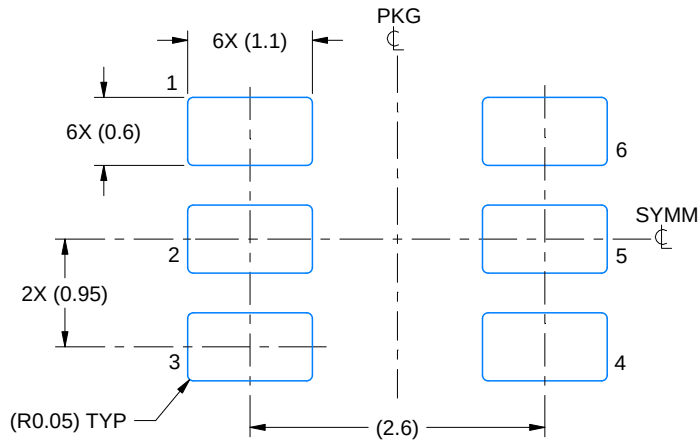
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.25 per side.
4. Leads 1,2,3 may be wider than leads 4,5,6 for package orientation.
5. Reference JEDEC MO-178.

EXAMPLE BOARD LAYOUT

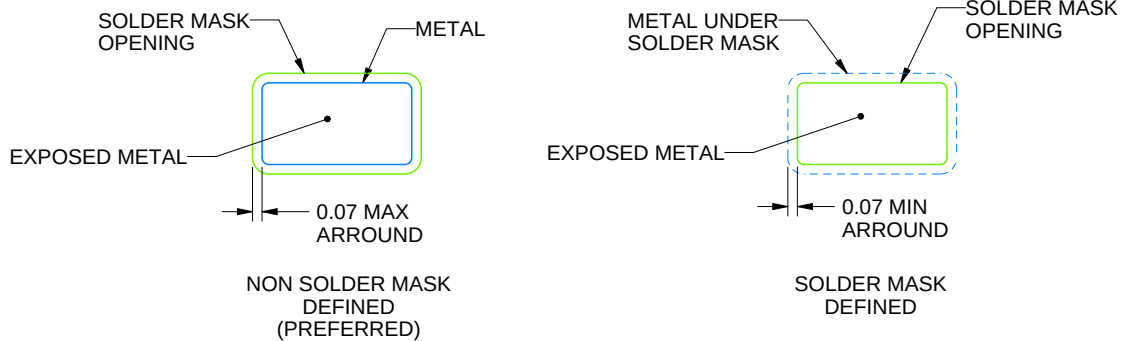
DBV0006A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:15X



SOLDER MASK DETAILS

4214840/E 02/2024

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

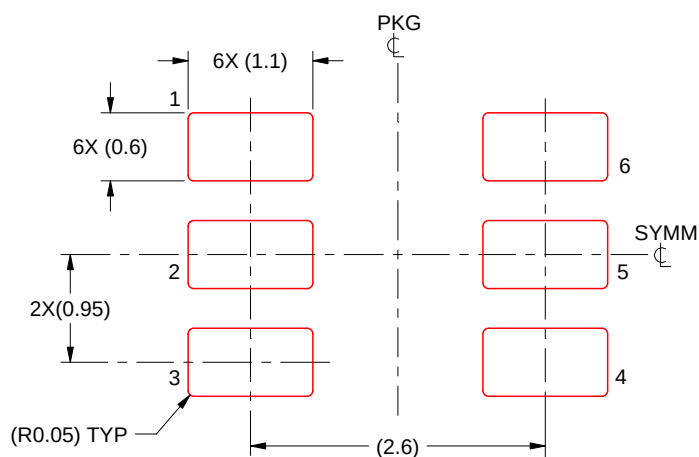
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DBV0006A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR

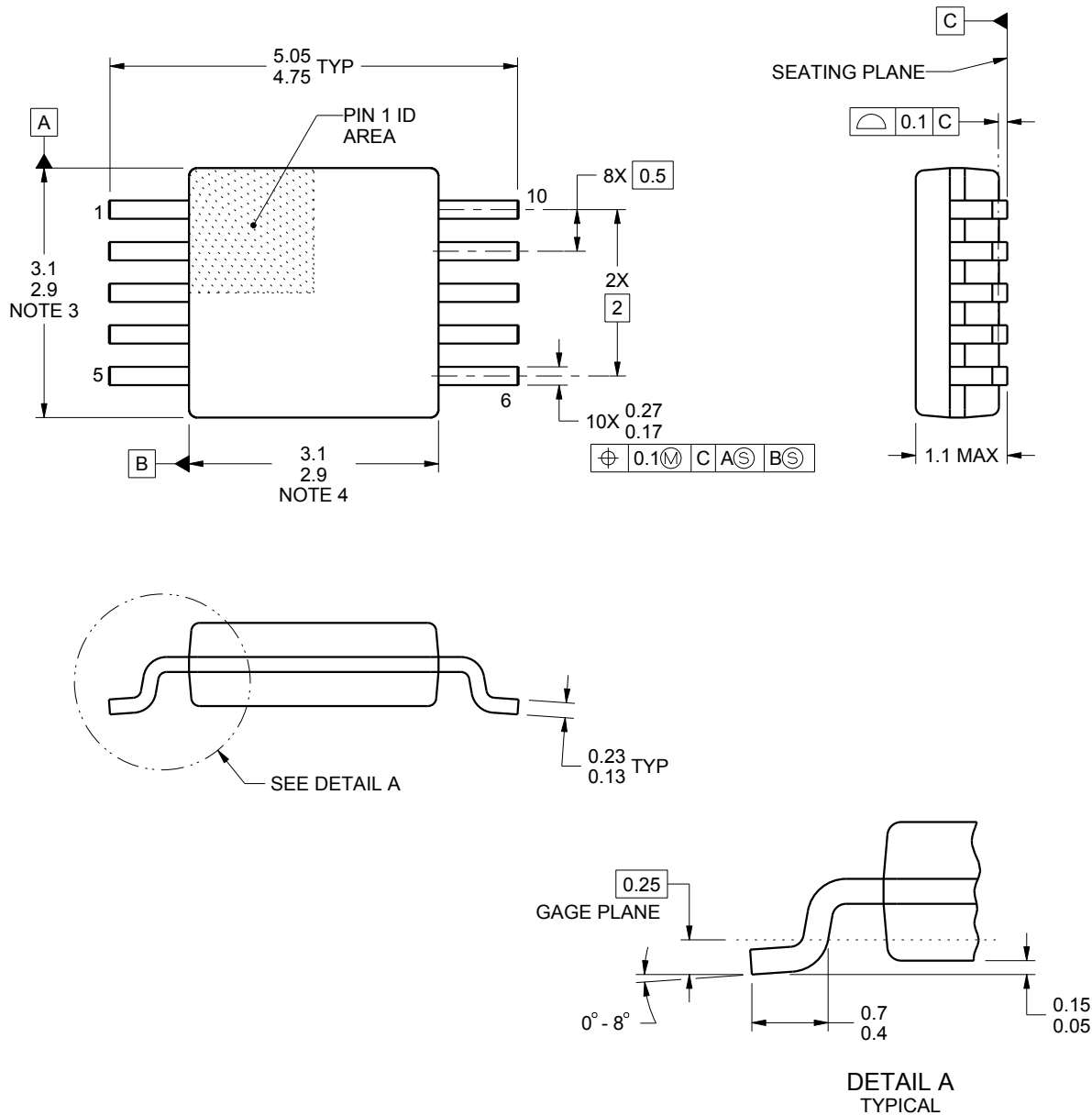


SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:15X

4214840/E 02/2024

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



4221984/A 05/2015

NOTES:

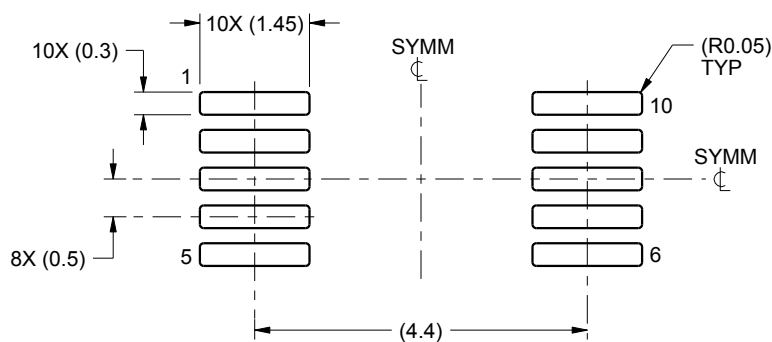
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187, variation BA.

EXAMPLE BOARD LAYOUT

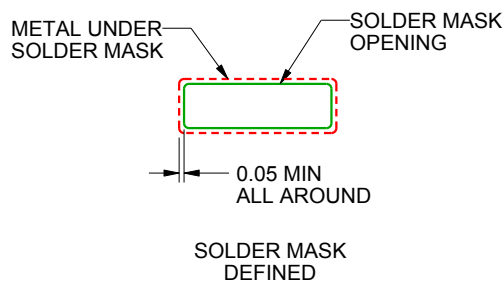
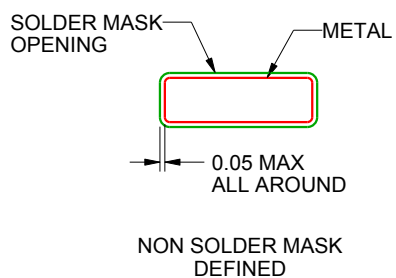
DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
SCALE:10X



SOLDER MASK DETAILS
NOT TO SCALE

4221984/A 05/2015

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

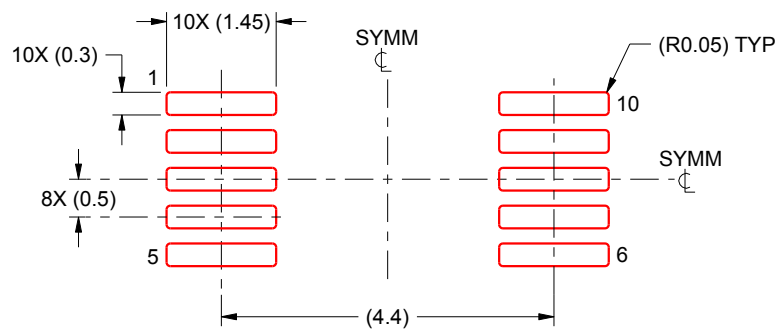
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:10X

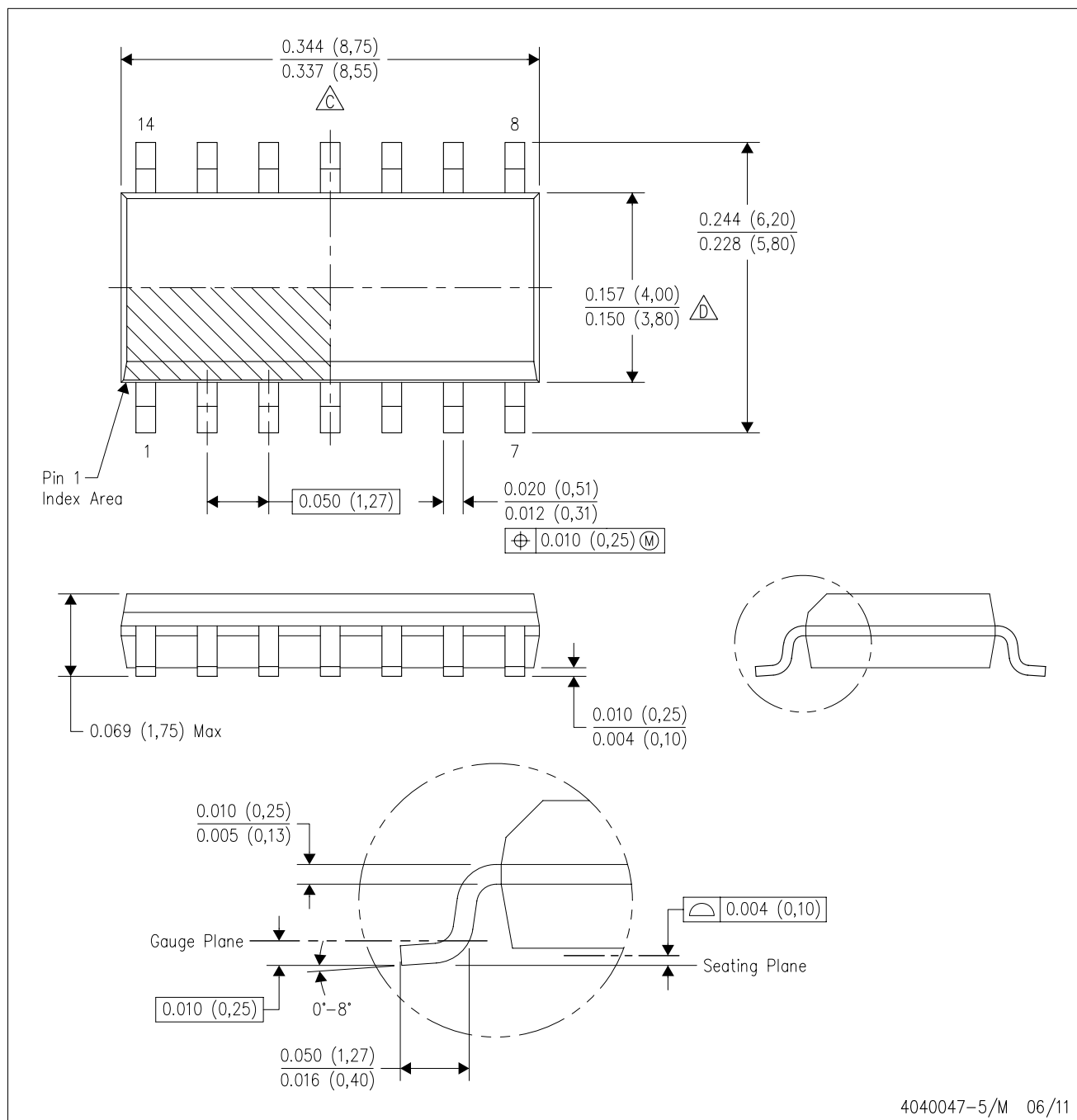
4221984/A 05/2015

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

D (R-PDSO-G14)

PLASTIC SMALL OUTLINE



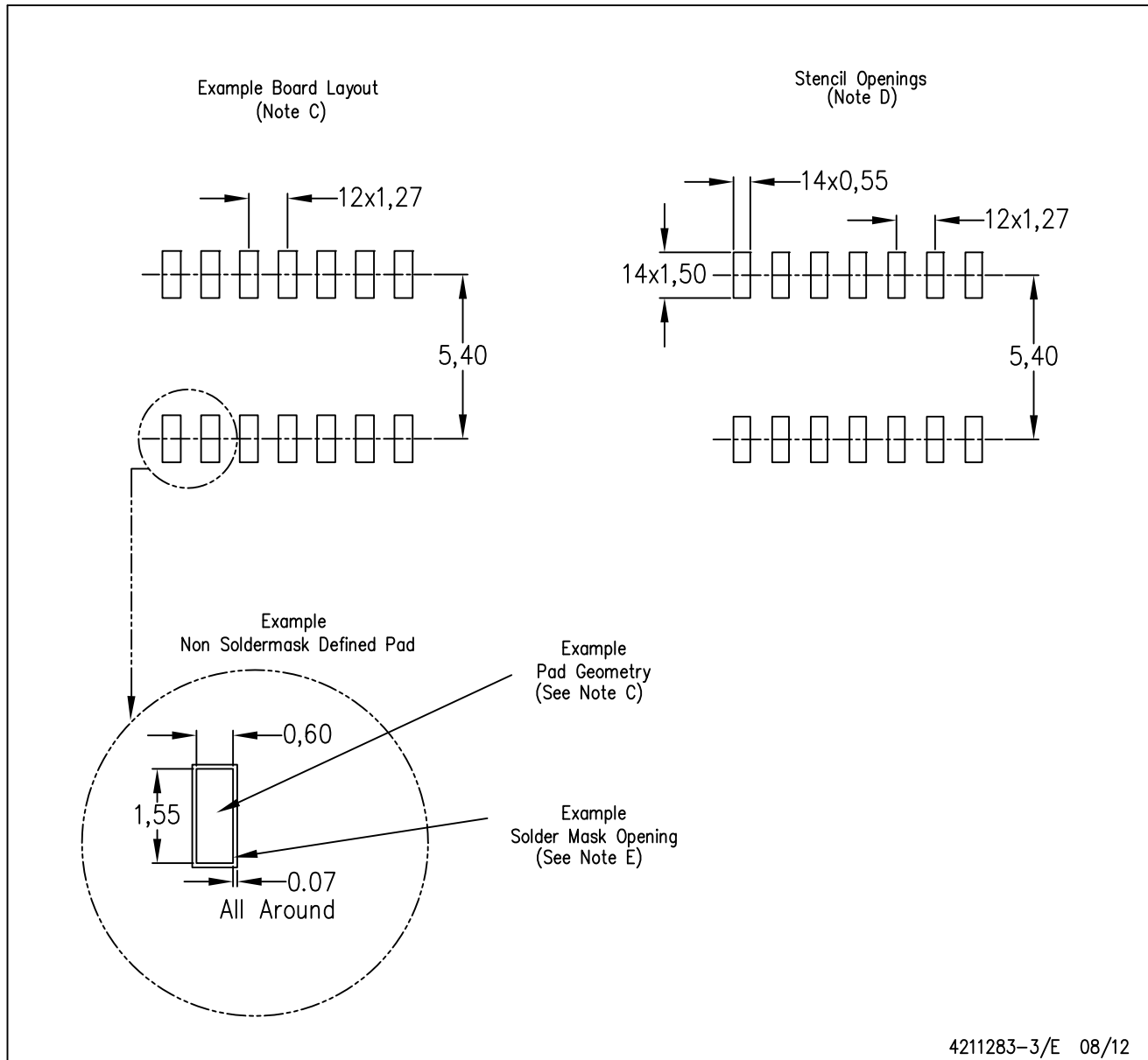
4040047-5/M 06/11

NOTES:

- A. All linear dimensions are in inches (millimeters).
- B. This drawing is subject to change without notice.
- C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
- D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
- E. Reference JEDEC MS-012 variation AB.

D (R-PDSO-G14)

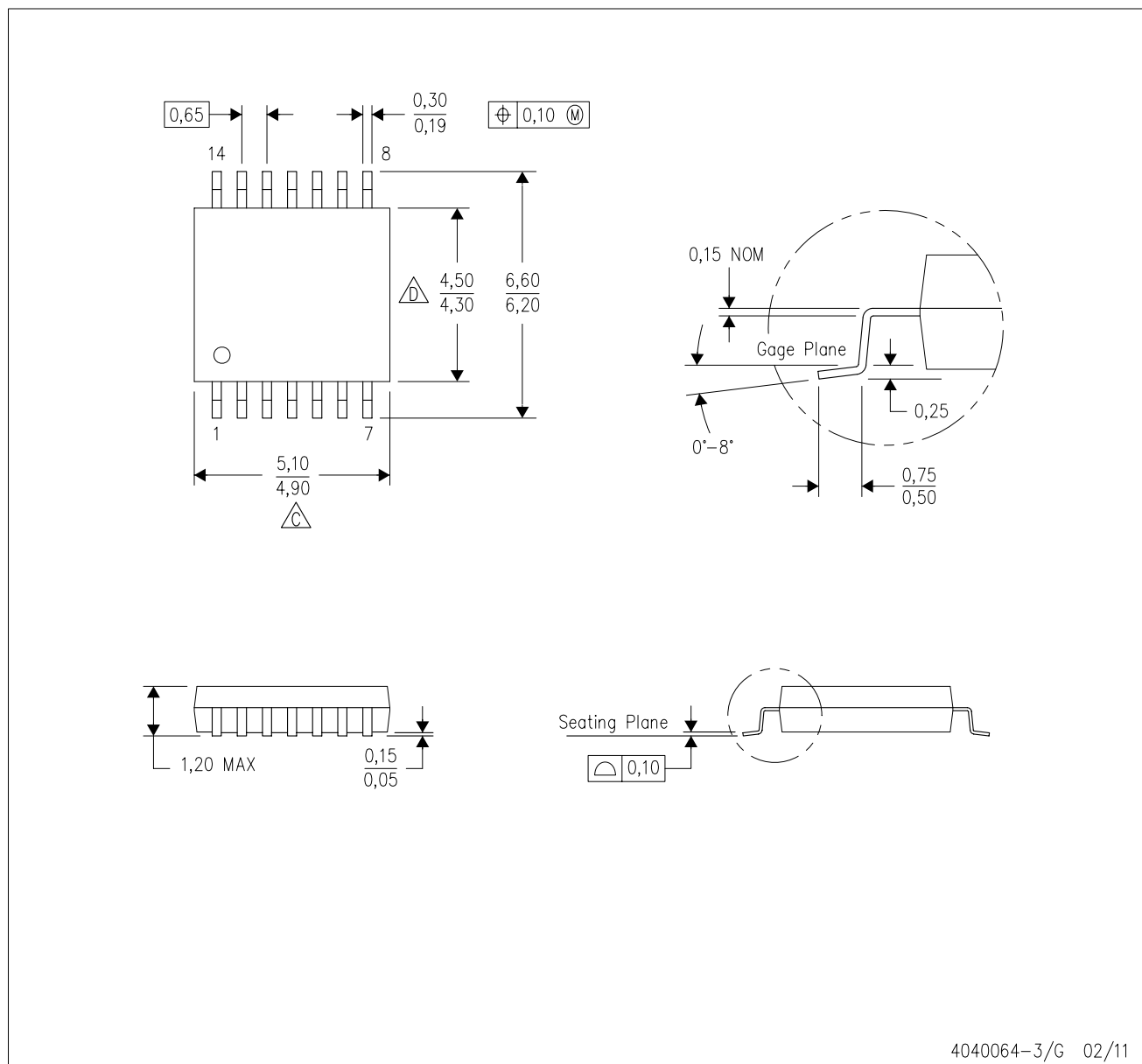
PLASTIC SMALL OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Publication IPC-7351 is recommended for alternate designs.
 - D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525 for other stencil recommendations.
 - E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PW (R-PDSO-G14)

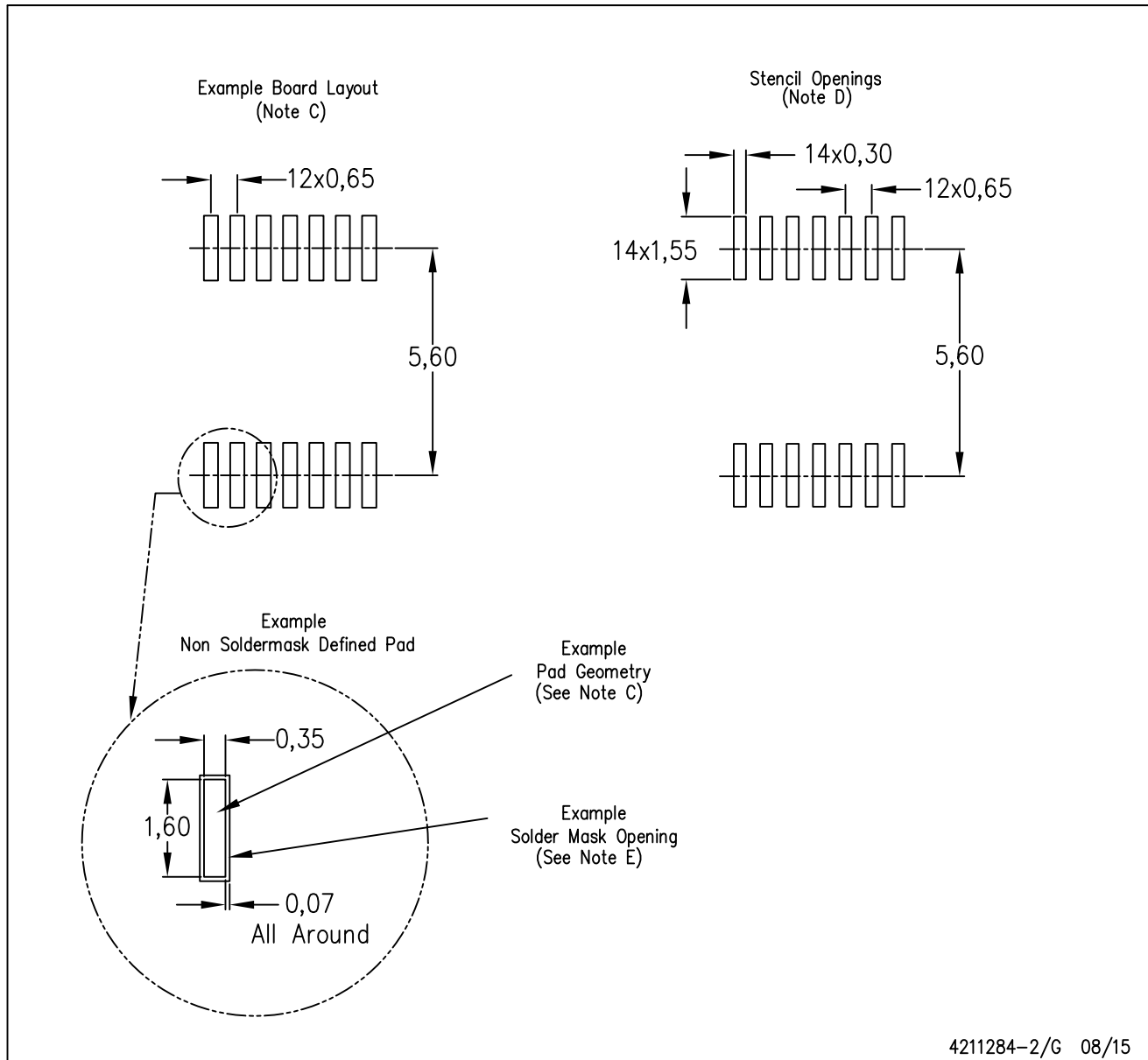
PLASTIC SMALL OUTLINE



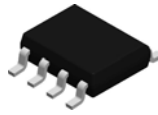
- NOTES:
- A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.
 - B. This drawing is subject to change without notice.
 - C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
 - D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
 - E. Falls within JEDEC MO-153

PW (R-PDSO-G14)

PLASTIC SMALL OUTLINE



- NOTES:
- All linear dimensions are in millimeters.
 - This drawing is subject to change without notice.
 - Publication IPC-7351 is recommended for alternate designs.
 - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525 for other stencil recommendations.
 - Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

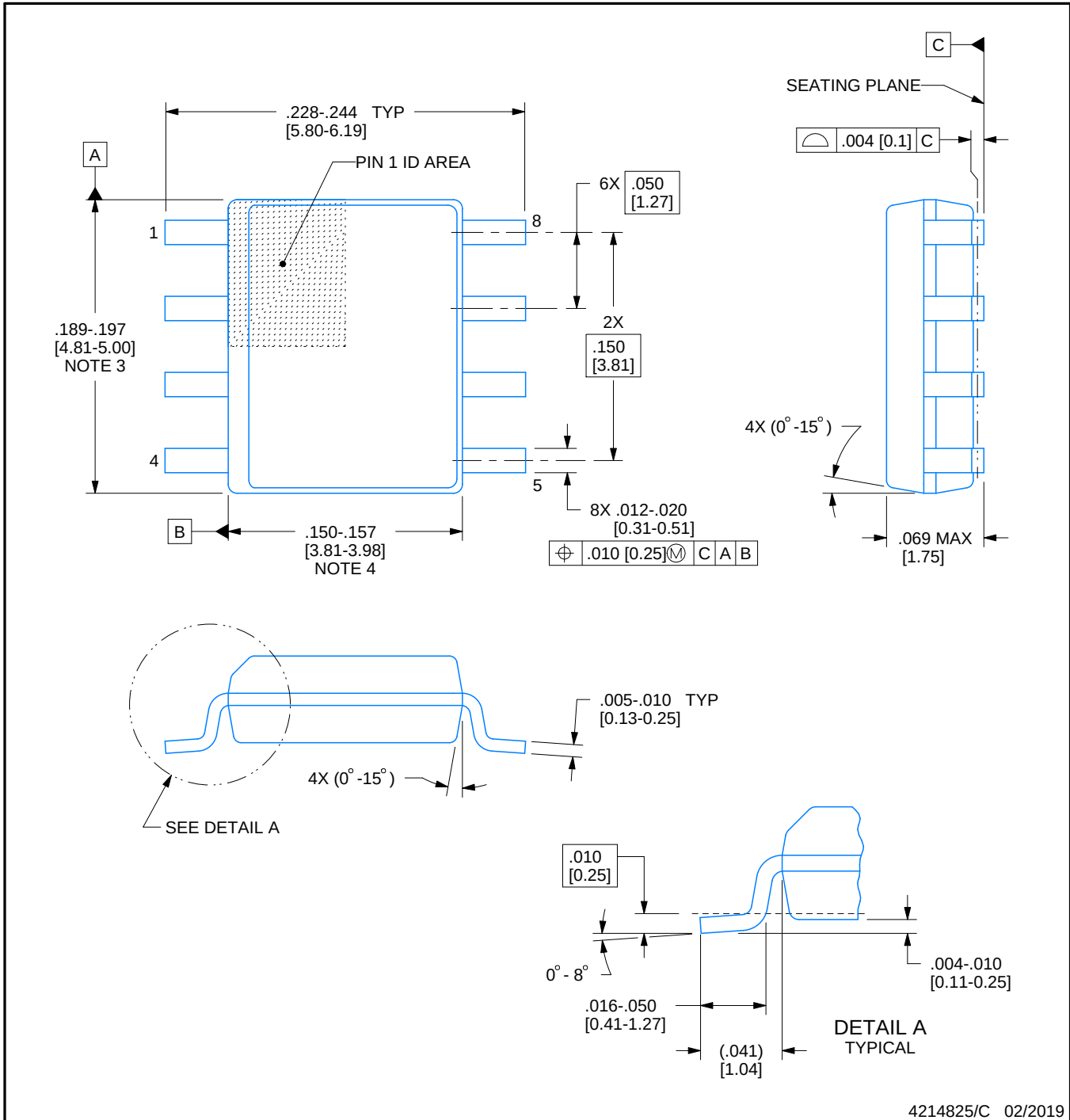


D0008A

PACKAGE OUTLINE

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

NOTES:

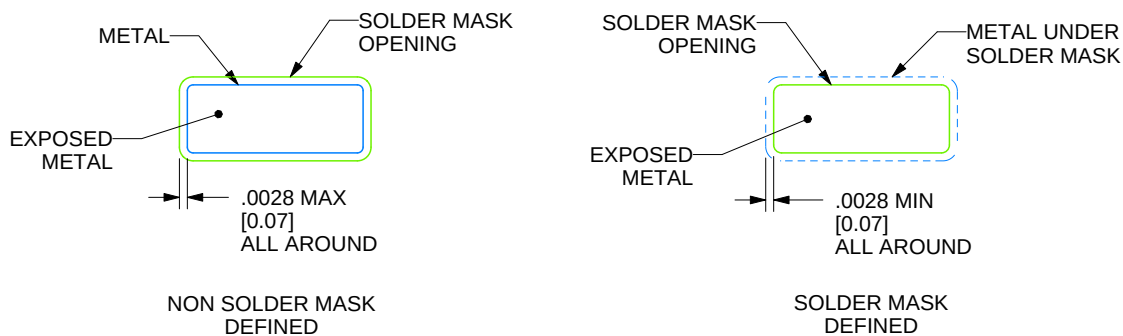
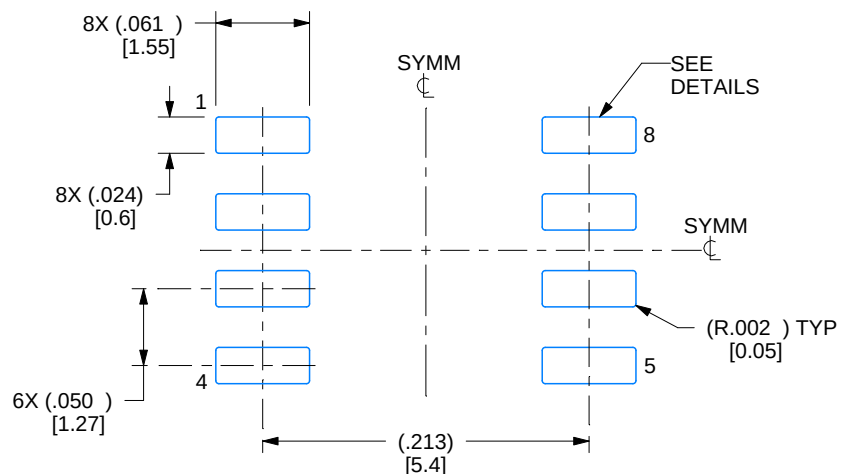
1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

EXAMPLE BOARD LAYOUT

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

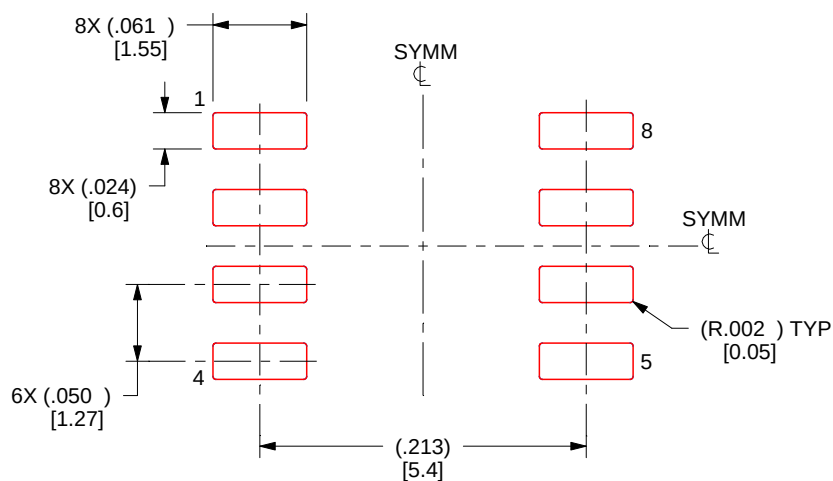
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

4214825/C 02/2019

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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